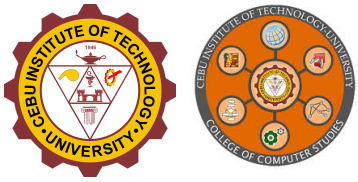
CEBU INSTITUTE OF TECHNOLOGY UNIVERSITY

COLLEGE OF COMPUTER STUDIES



Software Requirements Specification

for

ELEMENTOPIA

Submitted by:

2627-sem1-it411-53

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Date	Version	Description
May 6, 2026	1.0	Initial creation of SRS document.
May 13, 2026	1.1	Creating the introduction and overall description sections.
May 14, 2026	1.2	Creating the specific requirements section and diagrams.
May 15, 2026	1.3	Addition of other diagrams.
May 23, 2026	1.4	Modifications on diagrams.
Oct.15, 2026	2.0	Enhanced contents of the introduction and scope based on MVP validation findings. Added more references to cover used MVP Validation Framework models and knowledge bases.
Oct. 16, 2026	2.0	Improved overall description and interface requirements contents based on MVP validation findings. Added FR-1.6, FR-2.4, and FR-3.4

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1. Introduction

1.1. Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a comprehensive, authoritative description of the refactored system requirements for Elementopia, an interactive web-based gamified chemistry learning platform. This Version 2.0 specification refactors the baseline requirements based on empirical findings from the MVP validation cycle conducted with 65 secondary students and 4 licensed science educators.

This project extends the inherited Elementopia codebase by integrating three core functional modules: the Elemental Resonance Puzzle Module, the Assessment and Progress Module, and the Opponent-Based Challenge Module. The refactored specification formalizes the pedagogical scaffolding, real-time multiplayer resilience, and automated cloud telemetry necessary to ensure robust learning gains and dependable system deployment.

1.2. Scope

Elementopia targets secondary school science students learning chemistry concepts. The instructional scope is specifically bounded to the DepEd Grade 9 Science Curriculum (Matter and Chemical Bonding), covering atomic structures, valence electron sharing and transfer, Lewis structures, octet stability, and chemical formula synthesis.

This refactored functional scope encompasses:

- **Elemental Resonance Puzzle Module:** An environmental puzzle system where students synthesize chemical compounds on an interactive workbench to trigger reactions that clear physical obstacles. Features sequential room progression (Element Rooms to Compound Chambers), Meaningful Byproduct diagnostic alerts, the Hazmat Protocol failsafe (disabling invalid tiles upon 5 failures in 15 seconds), and the newly refactored Post-Synthesis Educational Debrief (displaying bond type, Lewis structure, and practical applications). *Each elemental domain is structured into three (3) scaffolded stage**s** of increasing pedagogical and stoichiometric difficulty: Stage 1 introduces foundational binary combinations, Stage 2 introduces multi-valence electron sharing, and Stage 3 introduces complex multi-atom or ionic compound synthesis under environmental obstacle pressure.*
- **Assessment and Progress Module:** Automated session telemetry logging (< 2 seconds) and a personalized visual Mastery Dashboard that highlights individual knowledge gaps. Refactored to include Centralized Diagnostic Pre-Test & Post-Test Synchronization to persist student cognitive baselines into Supabase.
- **Opponent-Based Challenge Module:** Peer-to-peer competitive matchmaking via auto-generated 5-digit room codes and real-time Speed-to-Compound winner evaluation. Refactored to incorporate a Realtime Heartbeat Disconnection Engine that gracefully manages network dropouts and auto-declares forfeit outcomes.

The platform is mobile-responsive and supports both touch-based and desktop

interaction. It does not include teacher administration tools or centralized classroom monitoring. The system does not require user registration; students are identified by self-assigned nicknames, and all session data is stored in Supabase.

1.3. Definitions, Acronyms and Abbreviations

- **SRS** — Software Requirements Specification; a document describing the functional and non-functional requirements of a software system.
- **Elementopia** — The proposed gamified web-based chemistry learning platform for high school students, extending an existing inherited system with three new modules.
- **MVP** — Minimum Viable Product; the core functional version of each module at a minimum of 60% functional completion.
- **UI/UX** — User Interface / User Experience; the visual design and interaction quality of the application.
- **Supabase** — An open-source backend platform providing PostgreSQL database storage and real-time data synchronization, used for all session data persistence and multiplayer communication in this system.
- **Elemental Resonance Domain** — A text-free environmental puzzle room in which a student synthesizes chemical compounds to trigger environmental reactions that clear physical obstacles blocking their progression.
- **Elemental Resonance** — The visual reaction animation triggered when a student successfully synthesizes the correct compound within a Domain, dissolving the obstacle and unlocking the path forward.
- **Meaningful Byproduct** -- A contextual, chemistry-based diagnostic message generated by the system for a predefined set of incorrect compound combination attempts, replacing binary "Wrong" notifications with feedback grounded in actual chemical properties.
- **Hazmat Protocol** — An automated failsafe triggered when the system detects five or more failed combination attempts within a 15-second sliding window, visually disabling irrelevant element tiles to narrow the student's focus without revealing the answer.
- **Element Room** — A foundational Domain puzzle room that must be cleared before a student can access Compound Chambers.
- **Compound Chamber** — An advanced Domain puzzle room that requires multi-step sequential compound synthesis and is only accessible after foundational Element Rooms are cleared.
- **Mastery Dashboard** — A personalized visual interface that renders a student's performance history from Supabase, displaying compound accuracy and completion speed per Domain to support self-directed knowledge gap identification.
- **Speed-to-Compound** — The competitive evaluation metric used in the

Opponent-Based Challenge Module that determines the match winner based on who reaches the correct compound synthesis first using the most efficient reaction path.

- **Access Code** — An auto-generated 5-digit alphanumeric code used to create or join an Opponent-Based Challenge room without manual configuration or user registration.
- **Session Nickname** — A temporary, self-assigned identifier entered by the student at the start of each session, used to associate performance data in Supabase without requiring formal authentication.
- **SRL** — Self-Regulated Learning; an educational framework in which learners independently monitor and control their own learning process.
- **Agile** — An iterative software development methodology supporting incremental delivery and continuous improvement.
- **Sprint** — A time-boxed development iteration in Agile methodology, typically lasting two to four weeks.
- **React** — The JavaScript frontend framework used to build the platform's user interface.
- **Spring Boot** — The Java-based backend framework used for API management and server-side logic.
- **Supabase Realtime** — The real-time data synchronization layer provided by Supabase, used to synchronize competitive match state between two players in the Opponent-Based Challenge Module.

1.4. References

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2.1. Product perspective

Elementopia is a web-based application that extends an existing gamified chemistry platform inherited from a prior development team. The three new modules—the Elemental Resonance Puzzle Module, the Assessment and Progress Module, and the Opponent-Based Challenge Module—are built on top of the original platform's features and are developed using React for the frontend, Spring Boot for backend logic and API management, and Supabase (PostgreSQL) for persistent session data storage and real-time multiplayer synchronization.

The system operates within a client-server architecture. The system processes puzzle logic, compound validation, and domain progression through a combination of React frontend logic and Spring Boot backend API management. Session performance data, including compound accuracy and completion speed, is written to and retrieved from Supabase, making all progress device-agnostic. The Opponent-Based Challenge Module relies on Supabase Realtime to synchronize match state between two competing players in real time.

The platform does not integrate with any external learning management system, school information system, or third-party authentication provider. It fills a specific gap in the science education technology ecosystem by combining three capabilities typically found in isolation: environmental simulation-based puzzle learning, automated behavioral analytics via Supabase, and peer-to-peer competitive challenge, all within a zero-registration, mobile-first environment.

Elementopia operates as a three-tier web application consisting of a React.js frontend, a Spring Boot Java backend API, and a Supabase PostgreSQL persistence layer. The system extends an inherited sandbox architecture into an objective-driven, scaffolded puzzle progression engine. It functions in both online cloud-synchronized mode and offline-graceful degradation mode (via browser localStorage cache).

2.2. User characteristics

The primary users are Grade 8 and Grade 9 High School Students learning chemical bonding. Users interact with the platform without mandatory registration, entering a temporary session nickname. Teachers and educators act as external Subject Matter Experts (SMEs), utilizing the system as a complementary laboratory pre-activity or flipped review aid.

General Learners (High School Students) use the platform as the intended following:

- Enter a temporary session nickname to access the platform without registration or authentication.
- Interact with Elemental Resonance Domain puzzle rooms by synthesizing chemical compounds to trigger environmental reactions that clear physical obstacles blocking their progression.

- Receive Meaningful Byproduct diagnostic feedback for incorrect compound combination attempt.
- Track personal performance through the visual Mastery Dashboard, which displays compound accuracy and completion speed per Domain retrieved from Supabase.
- Create or join Opponent-Based Challenge rooms using auto-generated 5-digit access codes to compete against peers in real-time Speed-to-Compound evaluation.

2.3. Constraints

The following constraints limit the design and operational deployment of the Elementopia system:

- **Curriculum Boundary:** The chemical reaction pool is strictly bounded to DepEd Grade 9 appropriate reactions (Water, Table Salt, Carbon Dioxide, Methane, Magnesium Oxide, Calcium Chloride, Rust).
- **Development Timeline:** The three new modules must be delivered across four Agile sprints. Feature complexity is scoped accordingly, with a minimum viable product threshold of 60% functional completion per module.
- **No Teacher Administration:** The platform intentionally omits centralized classroom monitoring, grade reporting, and instructor configuration tools. Evaluation is limited to session-based behavioral performance metrics logged to Supabase.
- **No User Registration:** The system does not implement any identity verification, OAuth, or third-party authentication. All user sessions are temporary and identified only by a self-assigned nickname. If a student changes their nickname, prior session data will not be associated with their new identity.
- **Network Dependency:** The Opponent-Based Challenge Module and the Assessment and Progress Module require a stable internet connection for Supabase data synchronization and Supabase Realtime communication. Network instability may disrupt match flow, scoring accuracy, and session logging.
- **Touch Device Optimization:** While the platform is designed to be fully mobile-responsive, rendering consistency may vary across different browser engines and operating system versions.
- **Inherited System Dependency:** The three new modules are built on top of the existing Elementopia platform. Any breaking changes to the original system's architecture may require corresponding adjustments to the new modules.

2.4. Assumptions and dependencies

The following assumptions underpin the requirements stated in this document. Changes to any of these factors may require corresponding revisions to the SRS:

- Target users possess devices with modern browser support (Chrome 110+, Firefox 100+, Safari 15+). The frontend requires a minimum screen resolution of 360px width. Network interactions assume a minimum bandwidth of 1 Mbps for uninterrupted multiplayer matches.
- The assumption that devices used by students support touch-based interaction or

standard mouse input, and have a minimum screen resolution of 360px width for proper rendering of the puzzle interface.

- The assumption that the Supabase project instance is properly configured with the required database schema and real-time channels prior to deployment.
- The assumption that all chemical compound content — element combinations, valid reaction outcomes, and Meaningful Byproduct diagnostic messages — will be curated and encoded into the predefined compound pool by the development team prior to deployment, based on the official high school chemistry curriculum.
- It is assumed that students will use the platform on their own personal devices for the duration of a session. Session data is stored in Supabase under the student's temporary nickname.
- The development of the three new modules depends on the stability of the inherited Elementopia codebase. The frontend depends on React, the backend depends on Spring Boot, and all session data persistence and real-time functionality depends on Supabase.

3.1. External interface requirements

3.1.1. Hardware interfaces

Elementopia is designed to run on any device equipped with a modern web browser. Compatible with desktop, laptop, tablet, and mobile devices. Touch targets must enforce minimum bounding dimensions of 48px by 48px to prevent touch input latency and mis-clicks. No specialized hardware peripherals are required. The minimum recommended screen resolution is 360px width to ensure proper rendering of the puzzle interface. The entire platform requires a stable network connection (Wi-Fi or mobile data) to maintain Supabase Realtime communication between two players.

3.1.2. Software interfaces

Elementopia operates entirely within a standard web browser and does not require installation of any additional software on the user's device. The system interfaces with the following:

- **Web Browser (Chrome v110+, Firefox v100+, Safari v15+)** - The primary runtime environment for all frontend rendering, puzzle interaction, and UI logic built with React.
- **Spring Boot (Java)** - The backend framework responsible for API management, compound validation logic, and server-side processing.
- **Supabase (PostgreSQL)** - Used to store and retrieve all session performance data, including compound accuracy and completion speed per Domain.
- **Supabase Realtime** - Used exclusively by the Opponent-Based Challenge Module to synchronize match state between two competing players in real time.
- **Browser Local Storage (HTML5 Web Storage API)** - Used as a client-side persistence and caching layer to store temporary session nicknames, offline telemetry queues, and pre-test diagnostic baselines for graceful degradation during network interruptions.
- **Spring Data JPA / Hibernate -** The object-relational mapping (ORM) framework within Spring Boot utilized to execute transactional database mutations to Supabase PostgreSQL.

3.1.3. Communications interfaces

The single-player modules (Elemental Resonance Puzzle Module and Assessment and Progress Module) require internet connectivity exclusively for reading and writing session data to Supabase. A minimum network bandwidth of 1 Mbps is recommended to ensure uninterrupted data synchronization. All communications between the frontend and Supabase

are conducted by the following:

- **HTTPS / TLS 1.3** - All communications between frontend and backend (Spring Boot / Supabase RESTful API endpoints) is encrypted over HTTPS, guaranteeing data integrity without transmitting personally identifiable information.
- **WebSocket Secure (WSS)** - Utilized by Supabase Realtime for persistent, bidirectional communication channels during multiplayer duels.

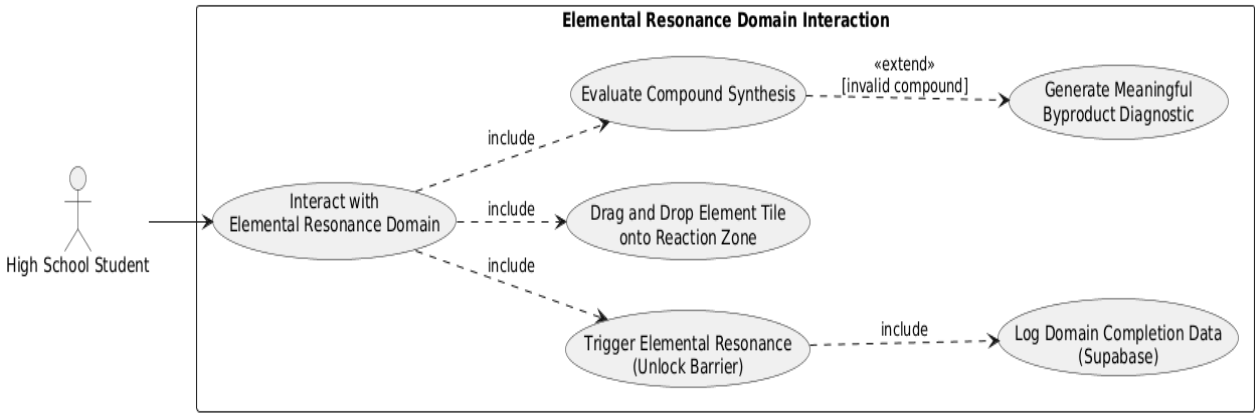
3.2. Functional requirements

Module 1: Elemental Resonance Puzzle Module

FR - 1.1 Elemental Resonance Domain Interaction

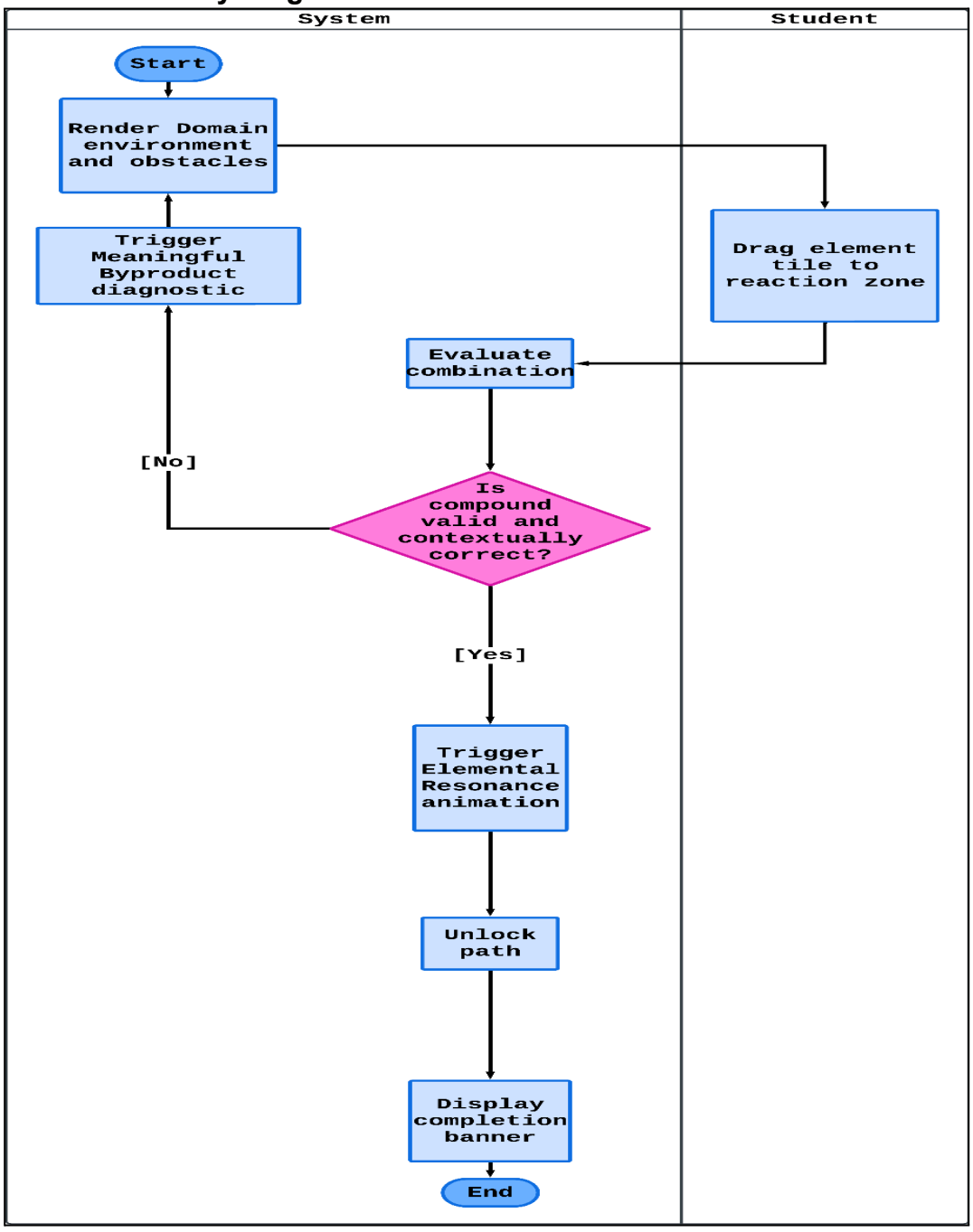
The system shall provide an interactive canvas workbench where students drag and drop chemical element tiles to synthesize compounds that neutralize environmental obstacles blocking room completion.

- **Use Case Diagram**

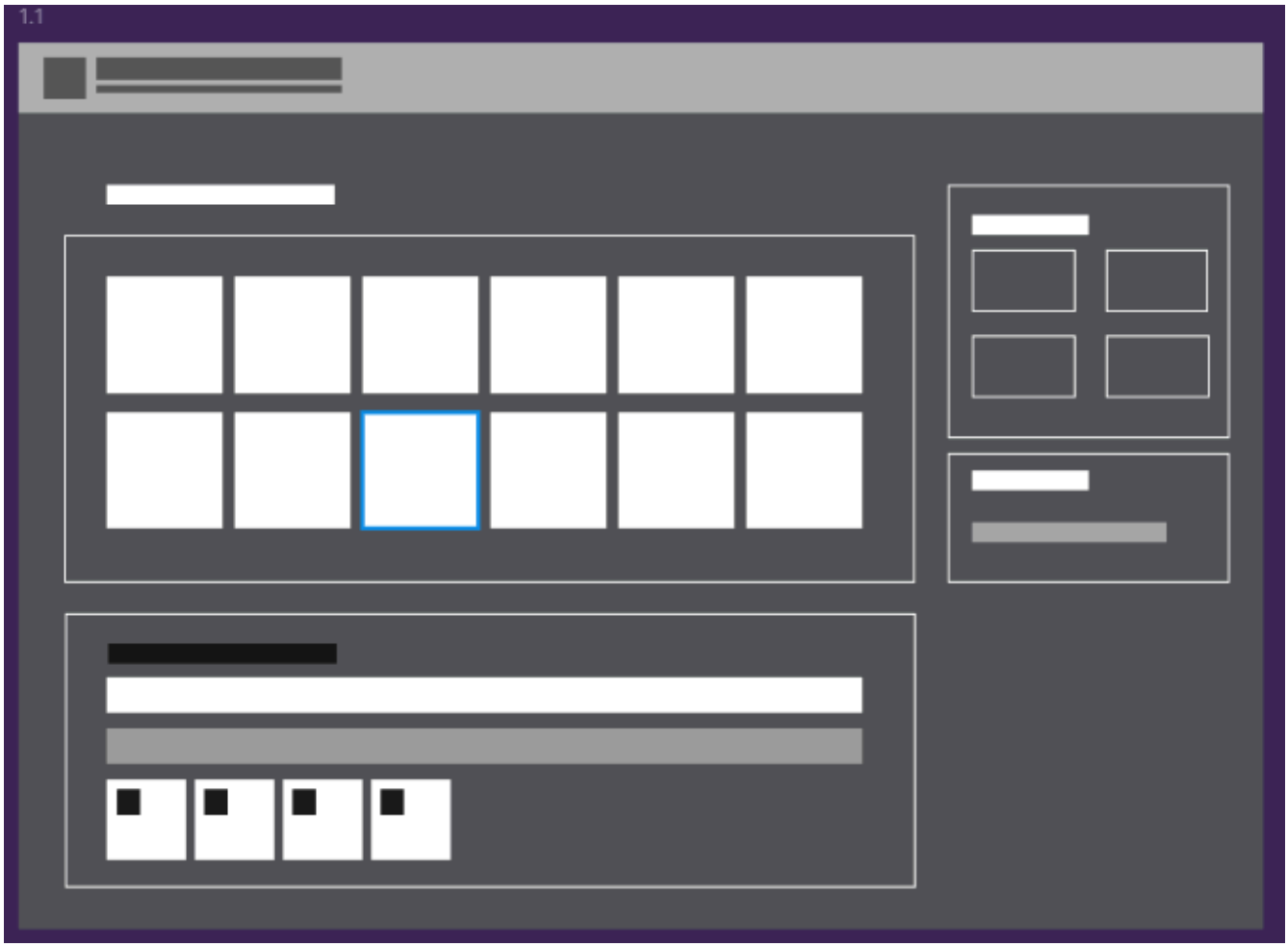


- **Use Case Description:** The student interacts with a text-free Elemental Resonance Domain environment. Rather than answering a quiz question, the student drags chemical element tiles onto the reaction zone to synthesize compounds and overcome a physical obstacle. A valid compound synthesis triggers a visual Elemental Resonance effect where the obstacle is cleared and the path forward is unlocked. An invalid combination generates a Meaningful Byproduct diagnostic message for a predefined set of wrong combinations, explaining why the attempt failed based on the chemical properties of the elements involved. Session completion data is logged to Supabase upon clearing the Domain.

- **Activity Diagram: Elemental Resonance Domain Interaction**



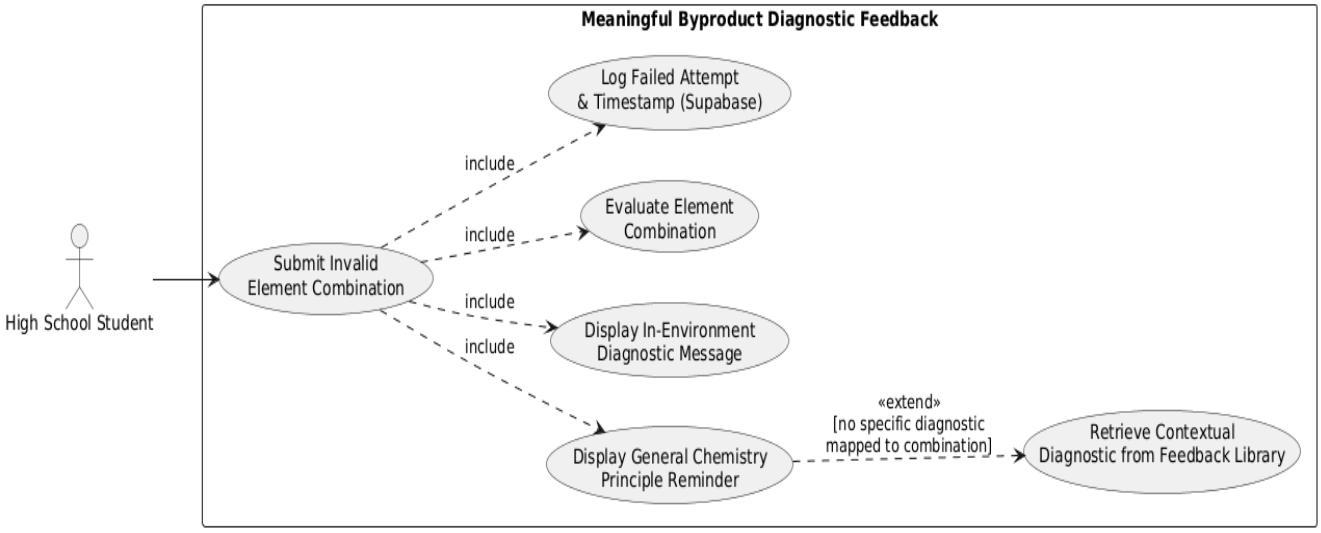
- Wireframe



FR - 1.2 Meaningful Byproduct Diagnostic Feedback

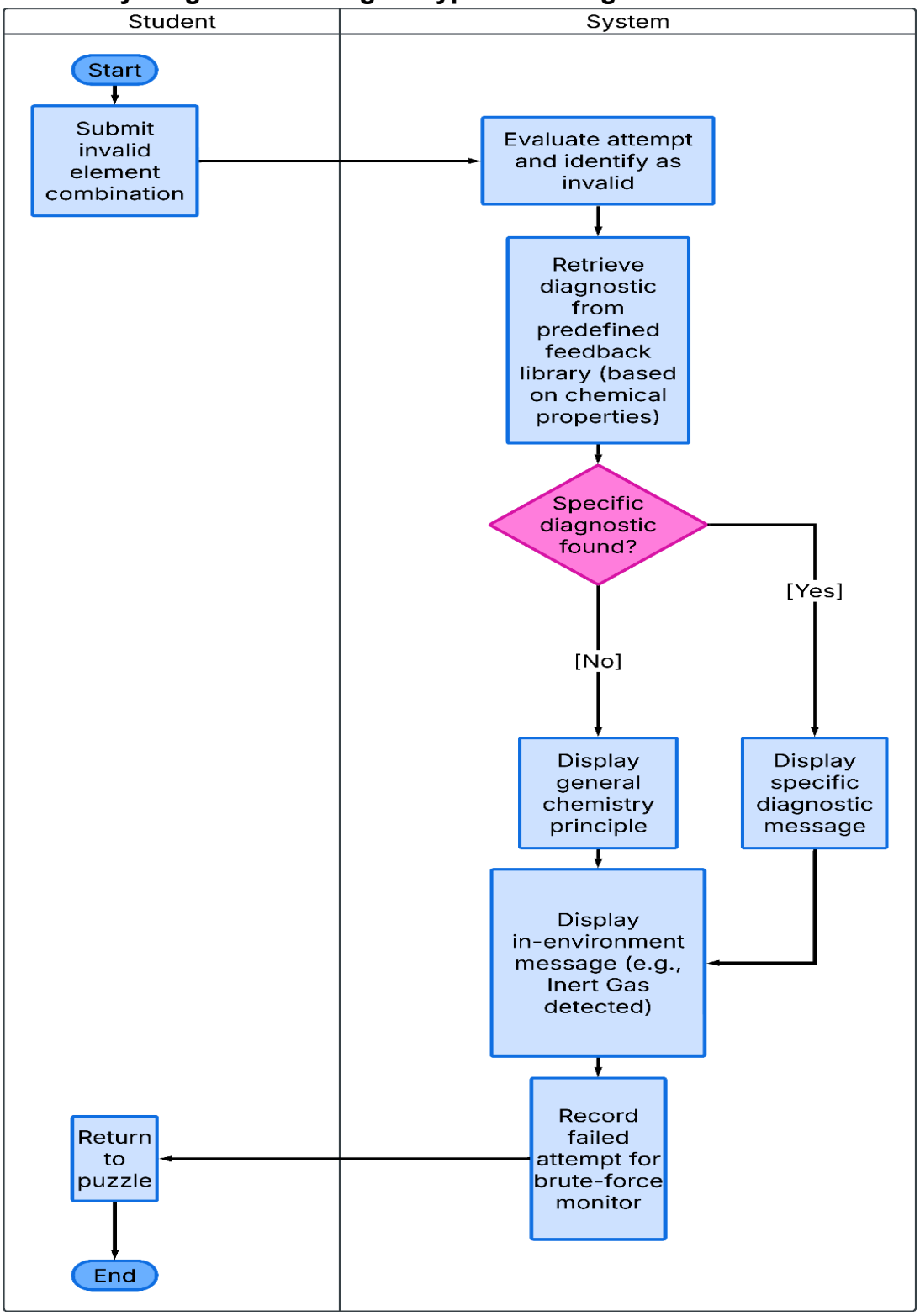
If an invalid reaction is attempted, the system shall evaluate the combination against a chemical property library and display a contextual 'Meaningful Byproduct' alert explaining octet or valence mismatch instead of a generic 'Wrong' notice.

- **Use Case Diagram**

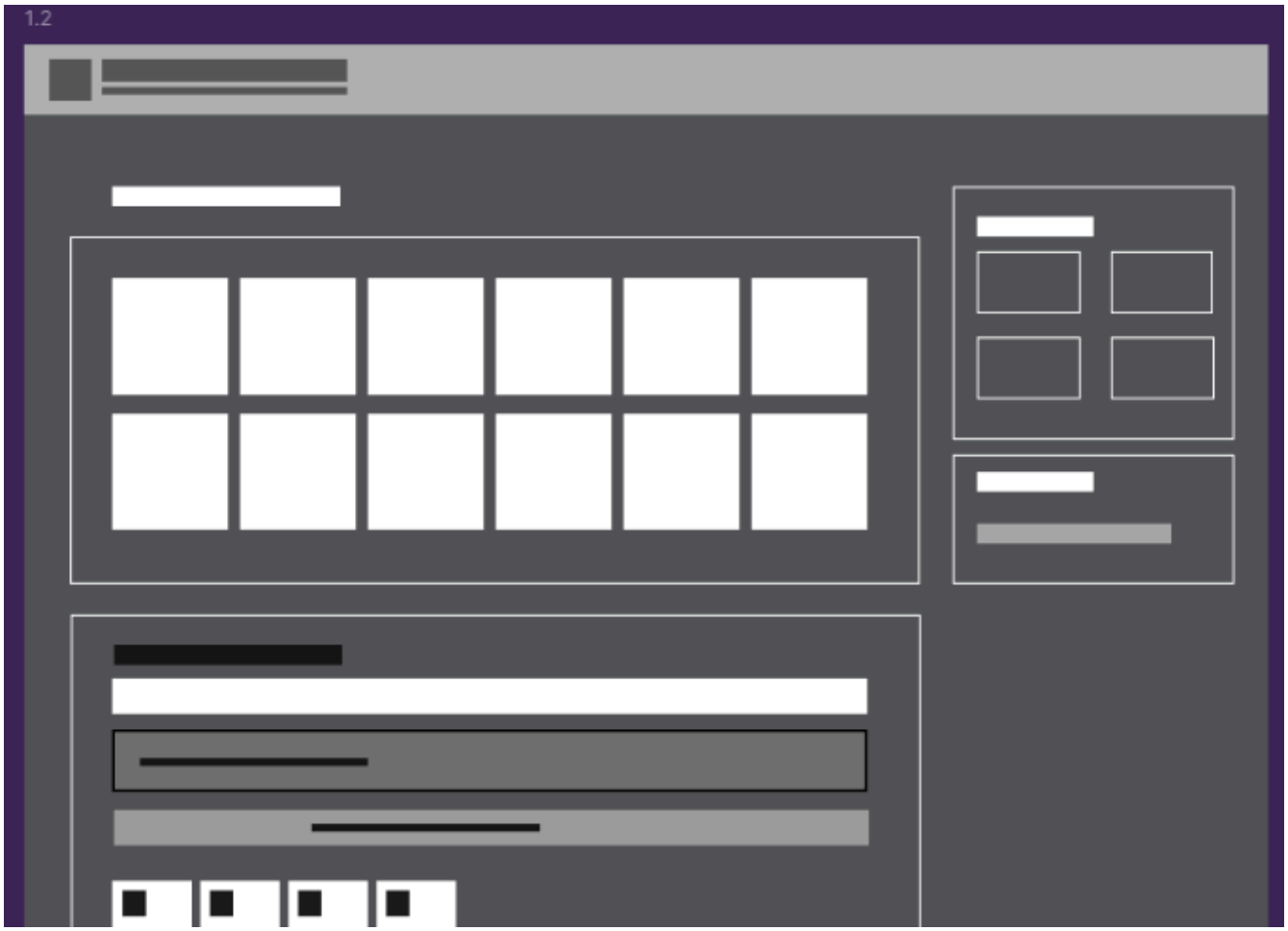


- **Use Case Description:** Upon an invalid compound combination attempt, the system checks the attempted combination against a predefined feedback library. If a specific diagnostic is mapped to that combination, the system displays a contextual, chemistry-based message explaining why the attempt failed based on the actual chemical properties of the elements involved. If no specific diagnostic is mapped, the system displays a general chemistry principle reminder. The failed attempt is recorded to support the Hazmat Protocol monitoring system.

- **Activity Diagram: Meaningful Byproduct Diagnostic Feedback**



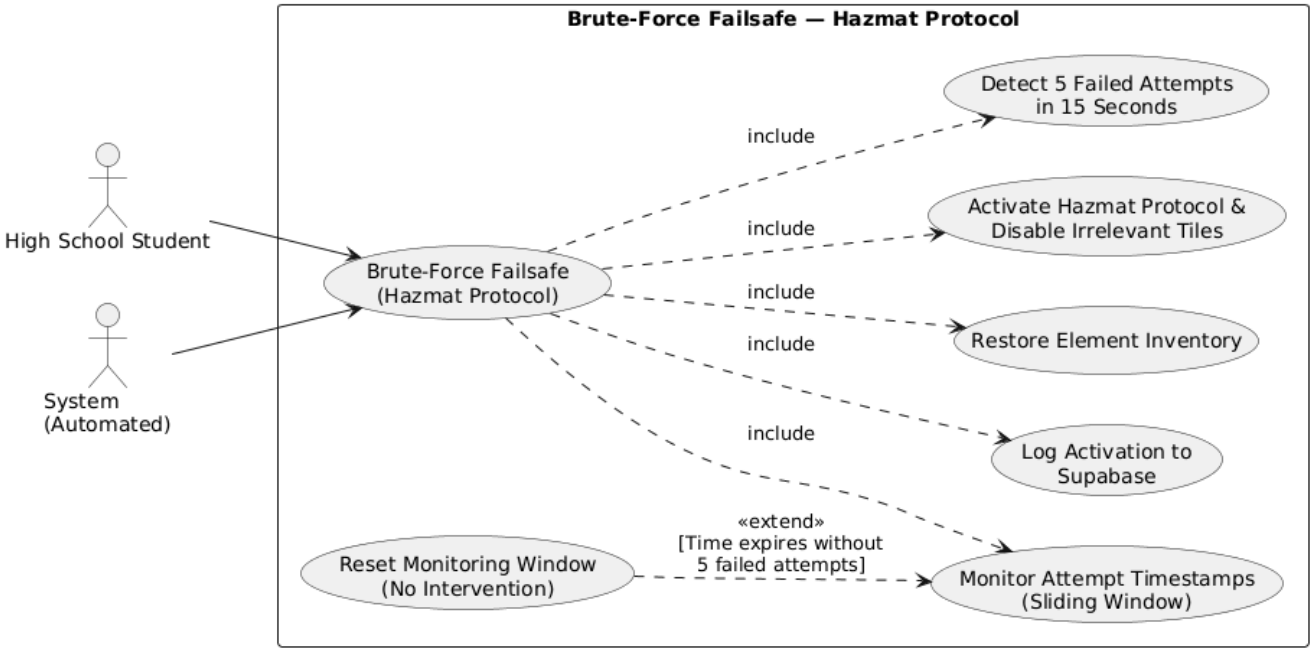
- **Wireframe**



FR - 1.3 Brute-Force Failsafe

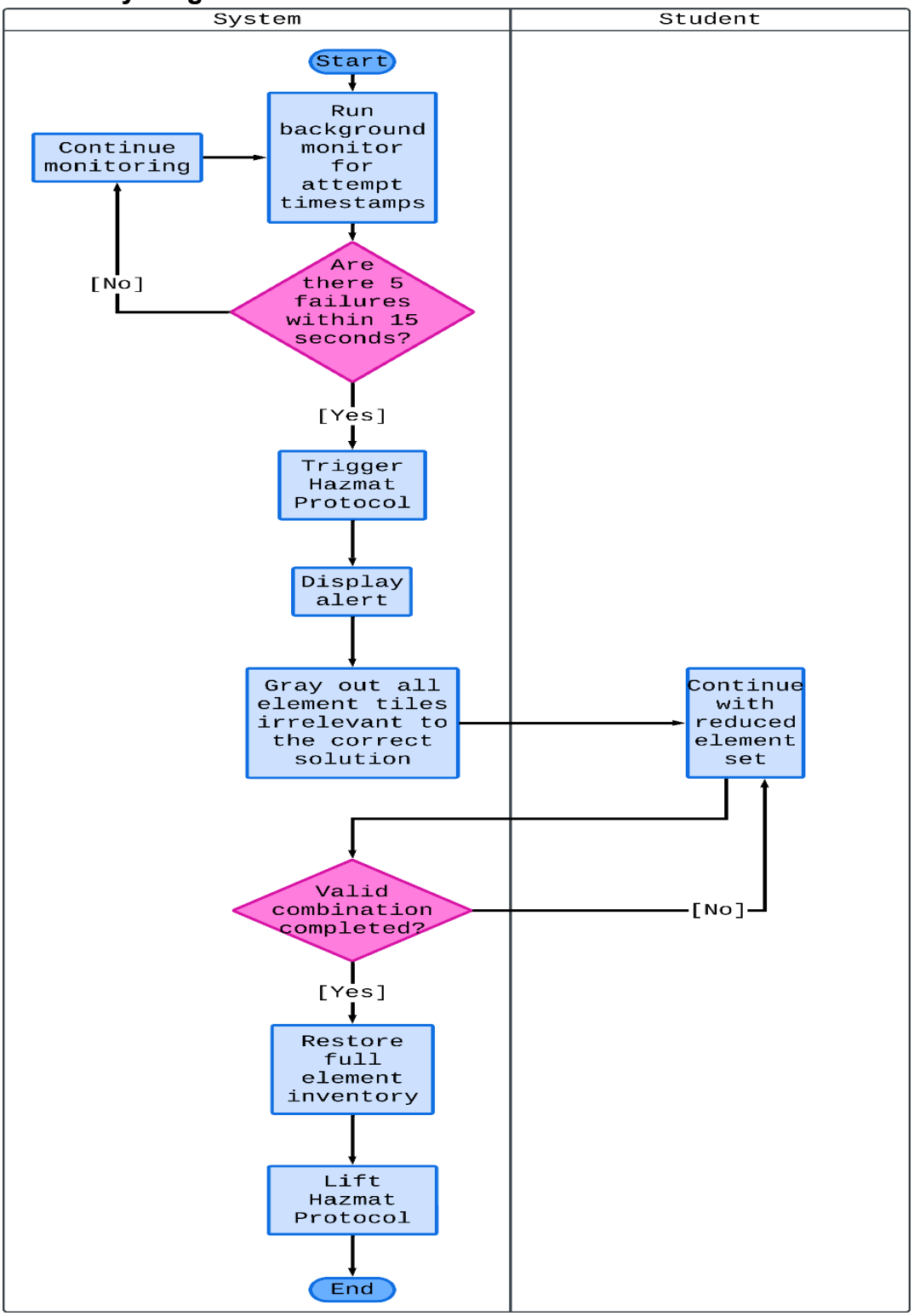
If a student commits 5 or more failed attempts within one stage of a domain, the Hazmat Protocol shall automatically gray-out and lock element tiles that do not belong to the target solution path.

- **Use Case Diagram**

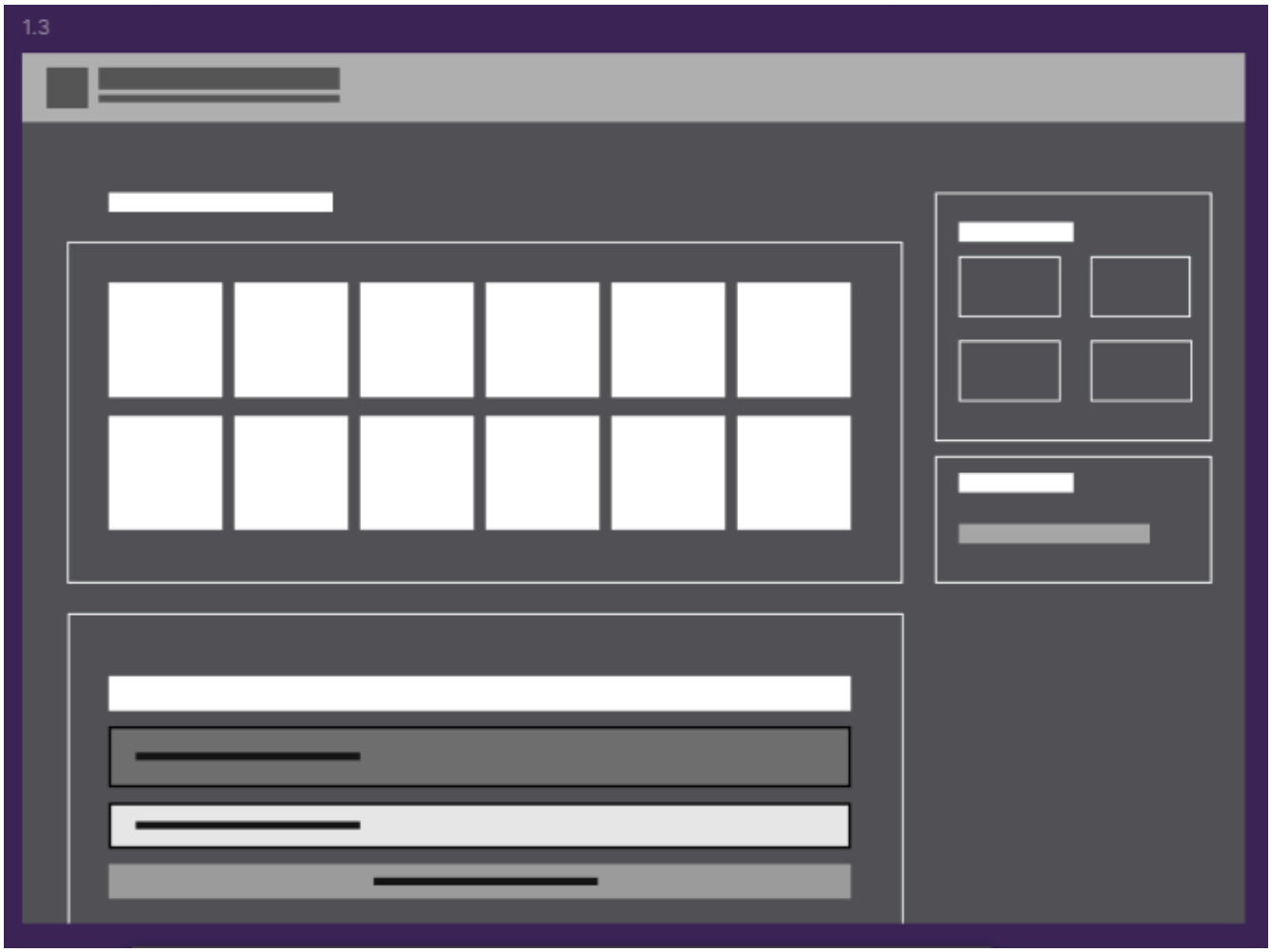


- **Use Case Description:** When the system detects five or more failed combination attempts within a 15-second sliding window, it automatically activates the Hazmat Protocol. This intervention visually disables element tiles that are irrelevant to the current puzzle, narrowing the student's focus without revealing the correct answer. Once the student successfully synthesizes a valid compound, the full element inventory is restored and the Hazmat Protocol is lifted. The activation is recorded to Supabase as part of the session performance log.

- **Activity Diagram: Brute-Force Failsafe**



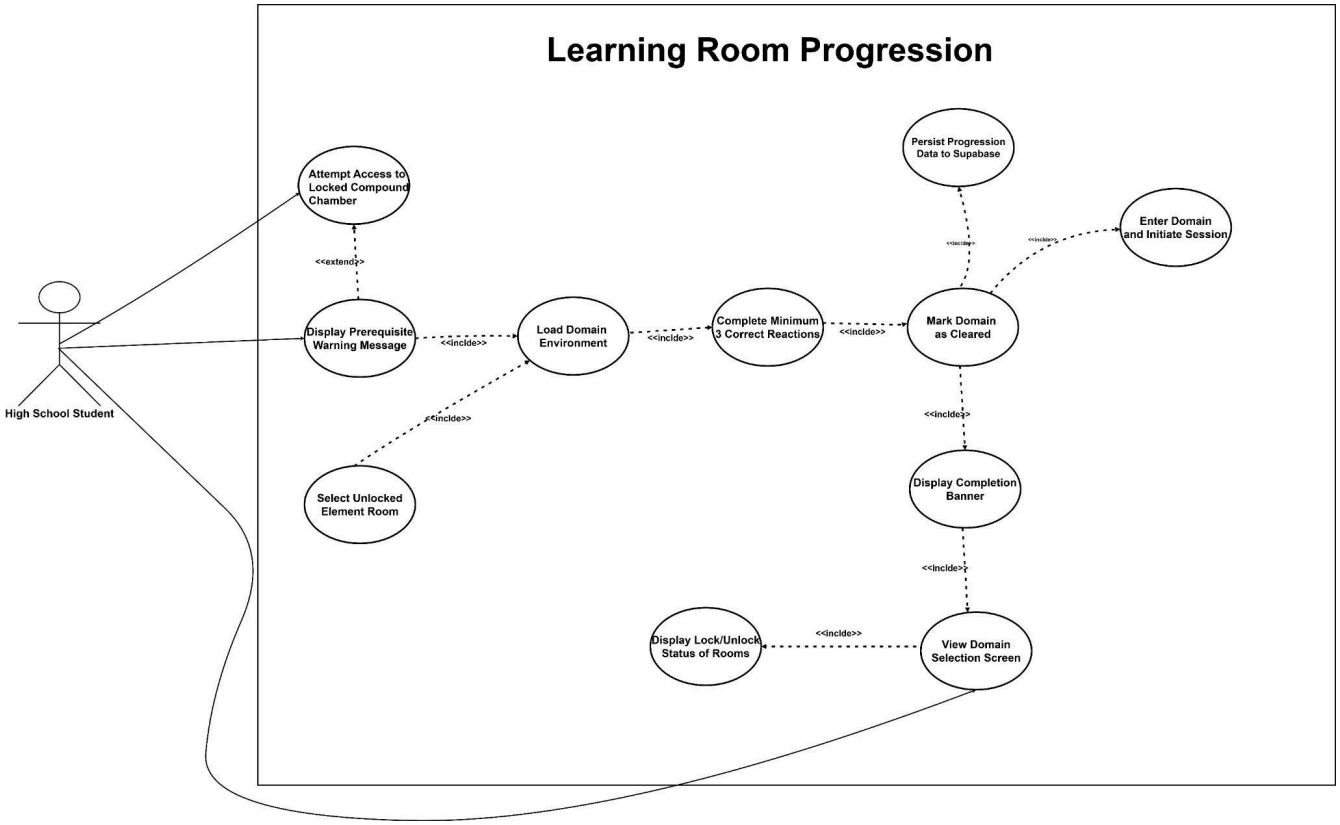
- **Wireframe**



FR - 1.4 Learning Room Progression

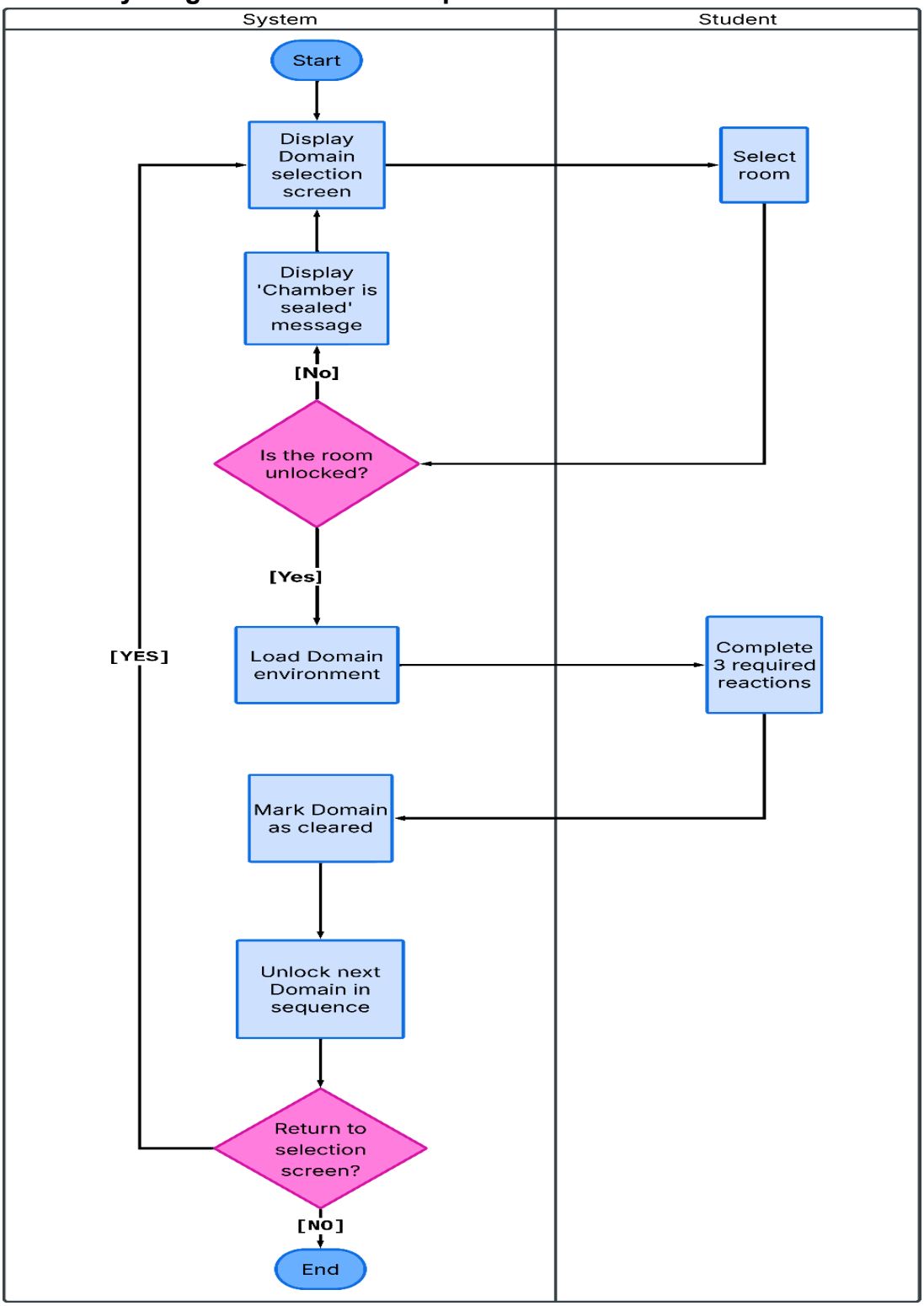
The system shall partition each Elemental Resonance domain into exactly three (3) sequential stages of ascending conceptual difficulty (Stage 1: Simple Binary Bonding; Stage 2: Stoichiometric Valence Ratios; Stage 3: Complex Molecular Synthesis). The system shall require the learner to clear each stage sequentially, locking progression to subsequent stages until the prerequisite reaction is validated and debriefed.

- **Use Case Diagram**

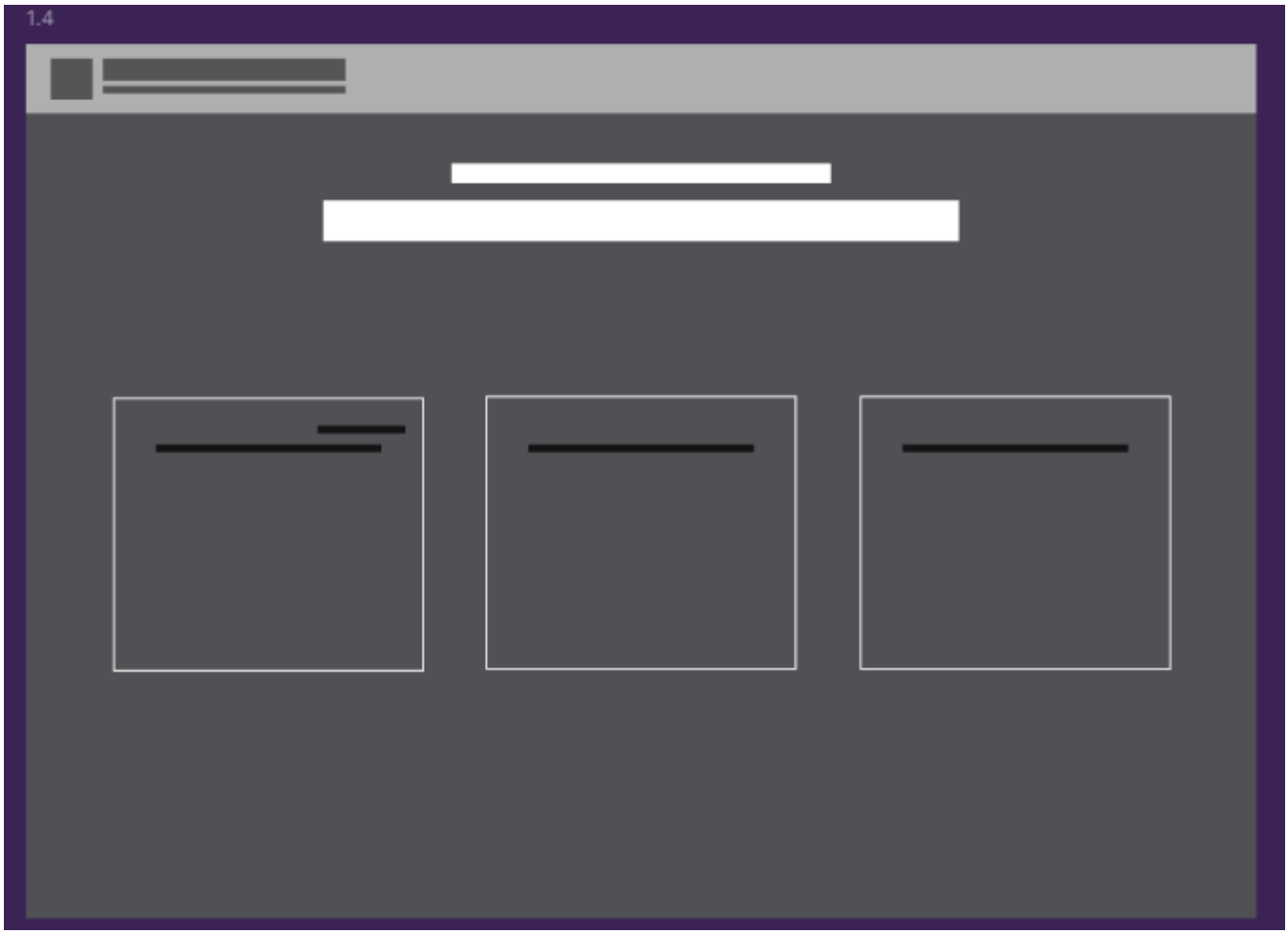


- **Use Case Description:** The system enforces a sequential unlocking structure for Domains. Students must clear foundational Element Rooms before accessing advanced Compound Chambers. A Domain is considered cleared when the student completes a minimum of three correct reactions within it. Upon clearing a Domain, the system marks it as complete, unlocks the next Domain in sequence, and displays a completion banner. Progression data is logged to Supabase so the student's unlock status persists across sessions.

- **Activity Diagram: Random Compound Generation Per Session**



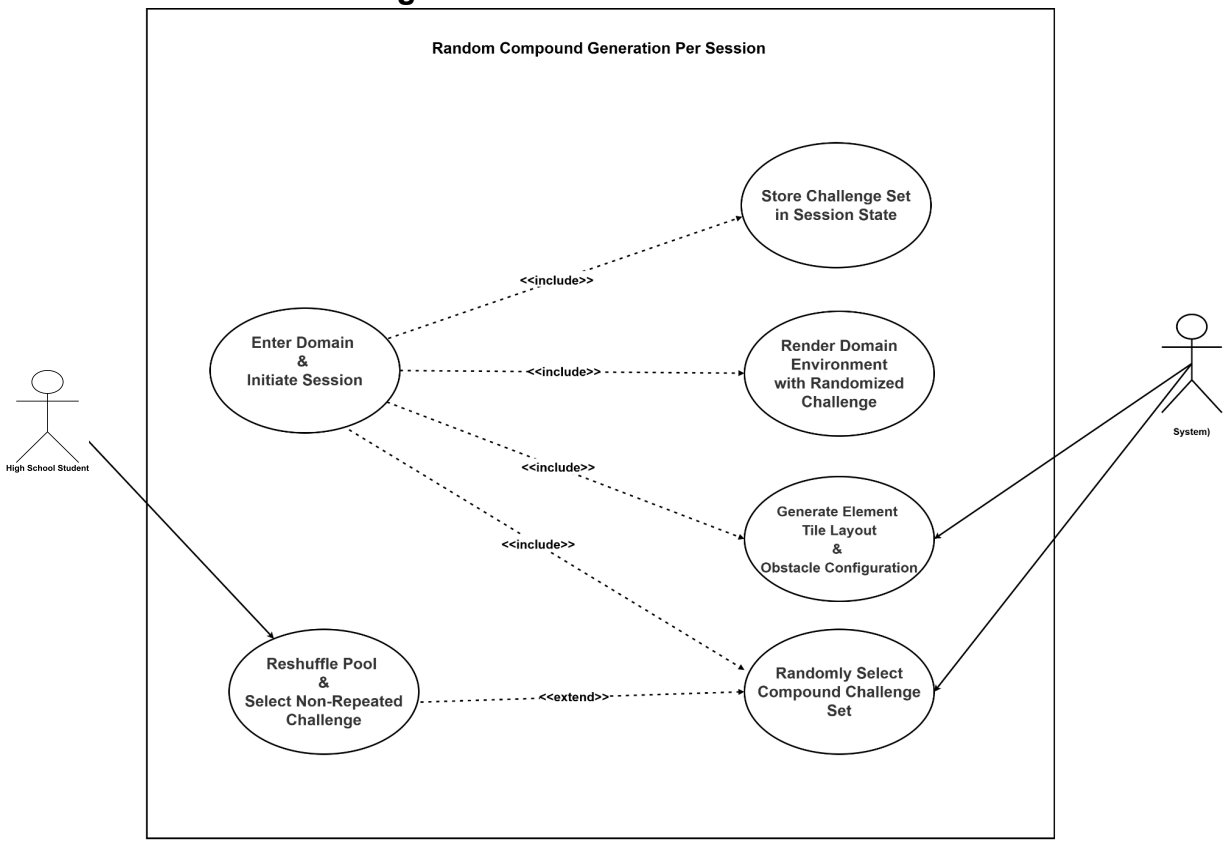
- **Wireframe**



FR - 1.5 Random Compound Generation Per Session

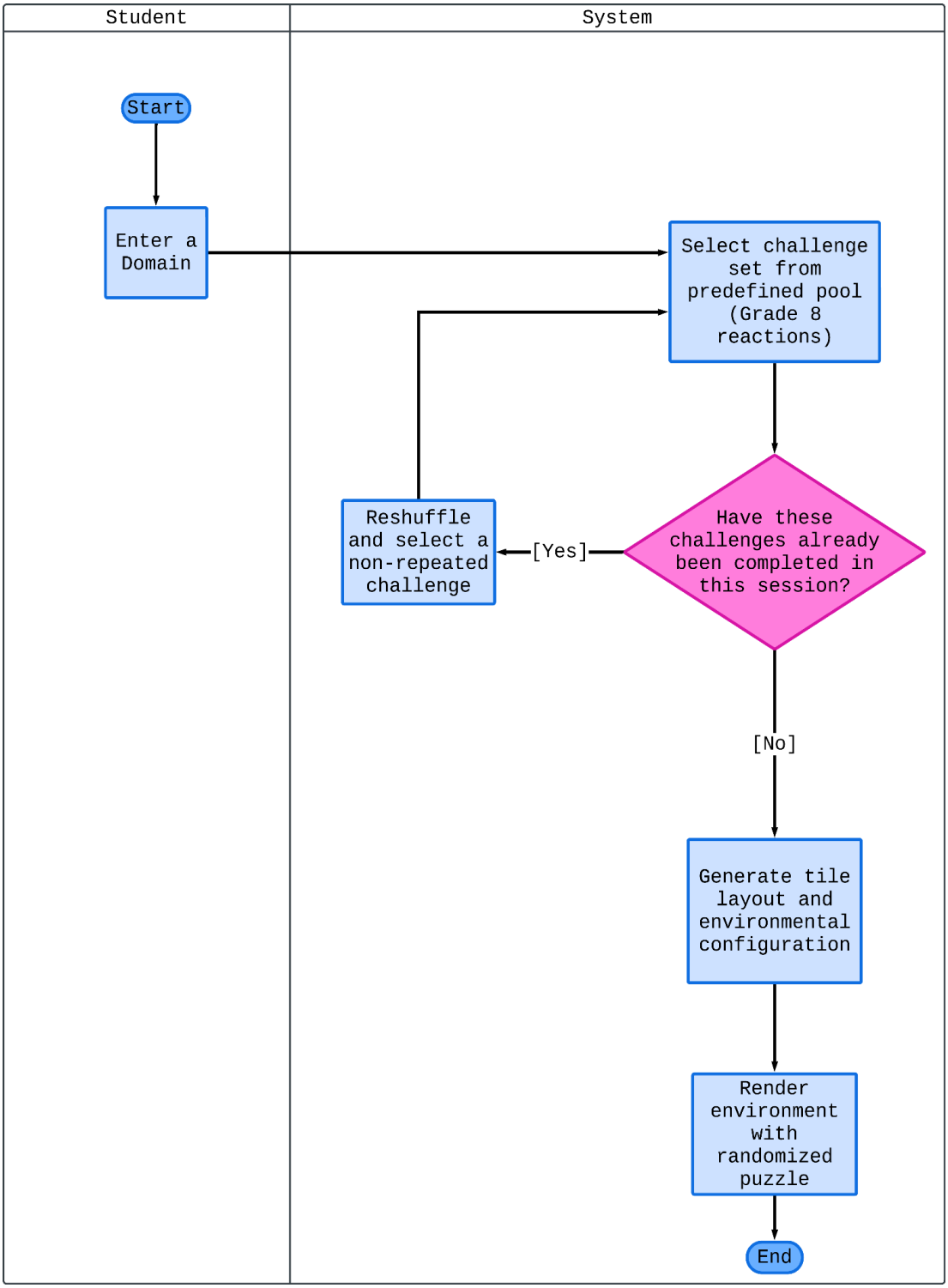
The system shall algorithmically randomize target compound assignments from the reaction pool at the start of each session to prevent rote memorization.

- **Use Case Diagram**



- **Use Case Description:** At the start of each Domain session, the system automatically selects a randomized set of chemical compound challenges from a predefined pool of high school chemistry reactions. The system checks whether the selected challenges have already been completed in the current session and reshuffles if necessary, ensuring students cannot rely on memorizing a fixed sequence of answers across sessions. The randomized challenge set is stored in session state for the duration of the Domain.

- **Activity Diagram: Random Compound Generation Per Session**



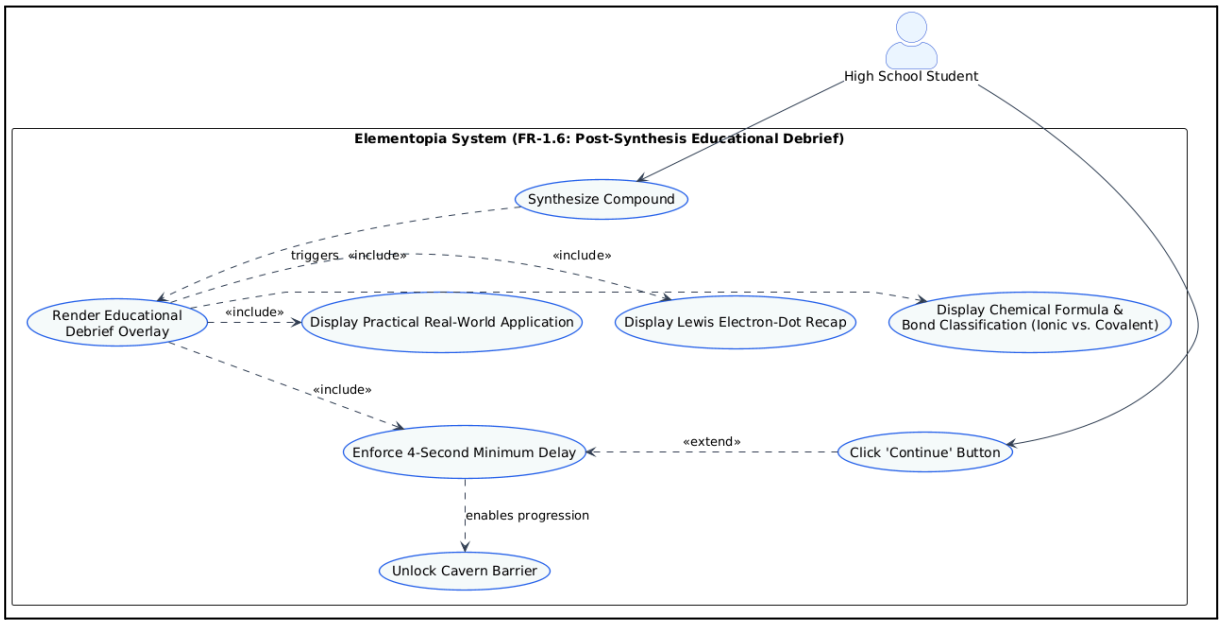
- **Wireframe**



FR-1.6: Post-Synthesis Educational Debrief

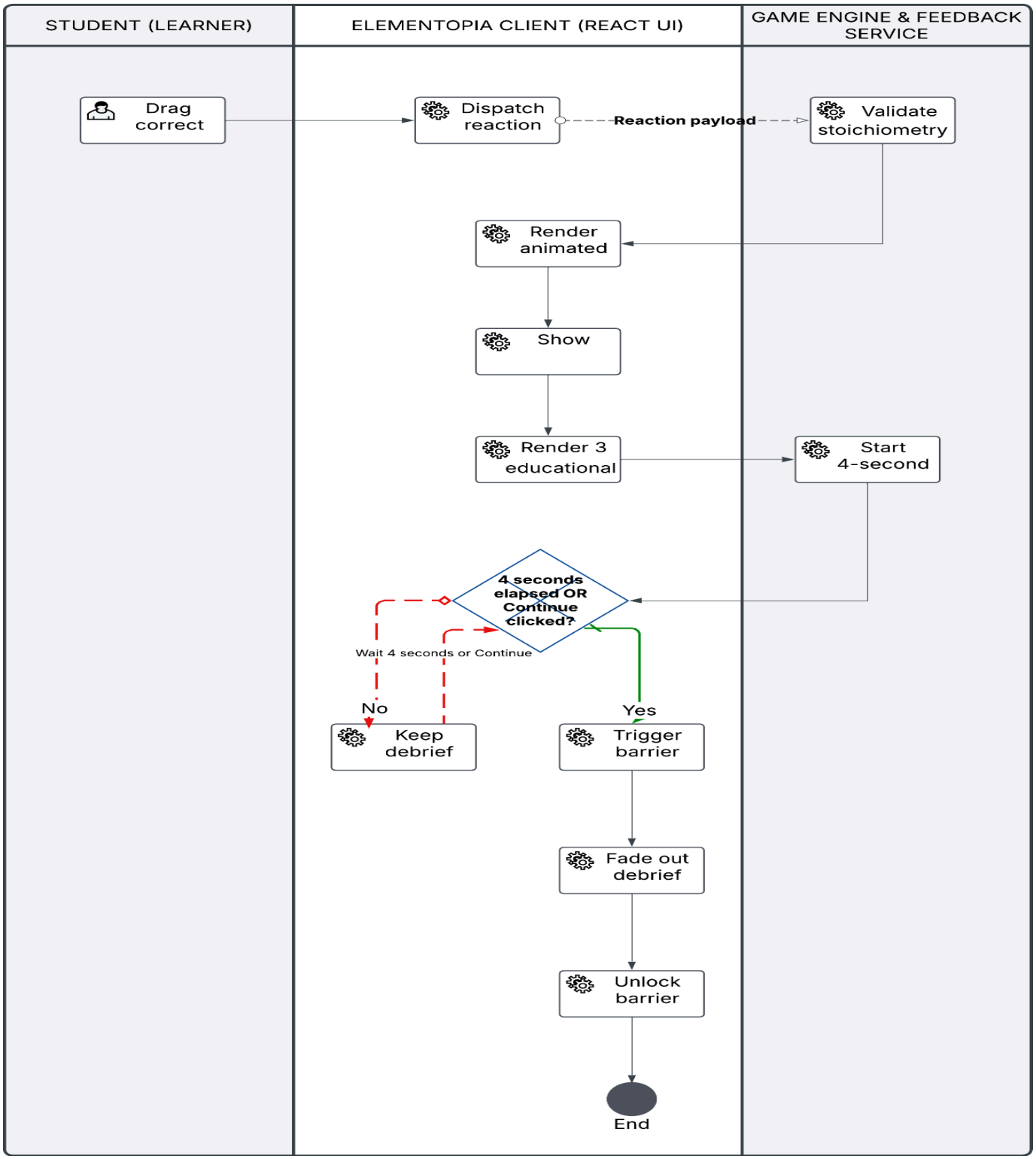
The system shall render an intermediate debrief displaying Doctor Atom's evaluation, the specific chemical bonding masteries unlocked (e.g., binary electron sharing, ternary multiple bonding rules), and a blueprint preview of the upcoming stage's target compound. The learner must click *"Dive Deeper to Stage {X}"* to proceed. After completion of one domain (stage 3), the system shall render a comprehensive mission summary displaying the learner's overall accuracy percentage, elapsed completion time, complete list of discovered compounds with chemical formulas, and a pedagogical review of the core chemical bonding concept mastered.

- **Use-Case Diagram**

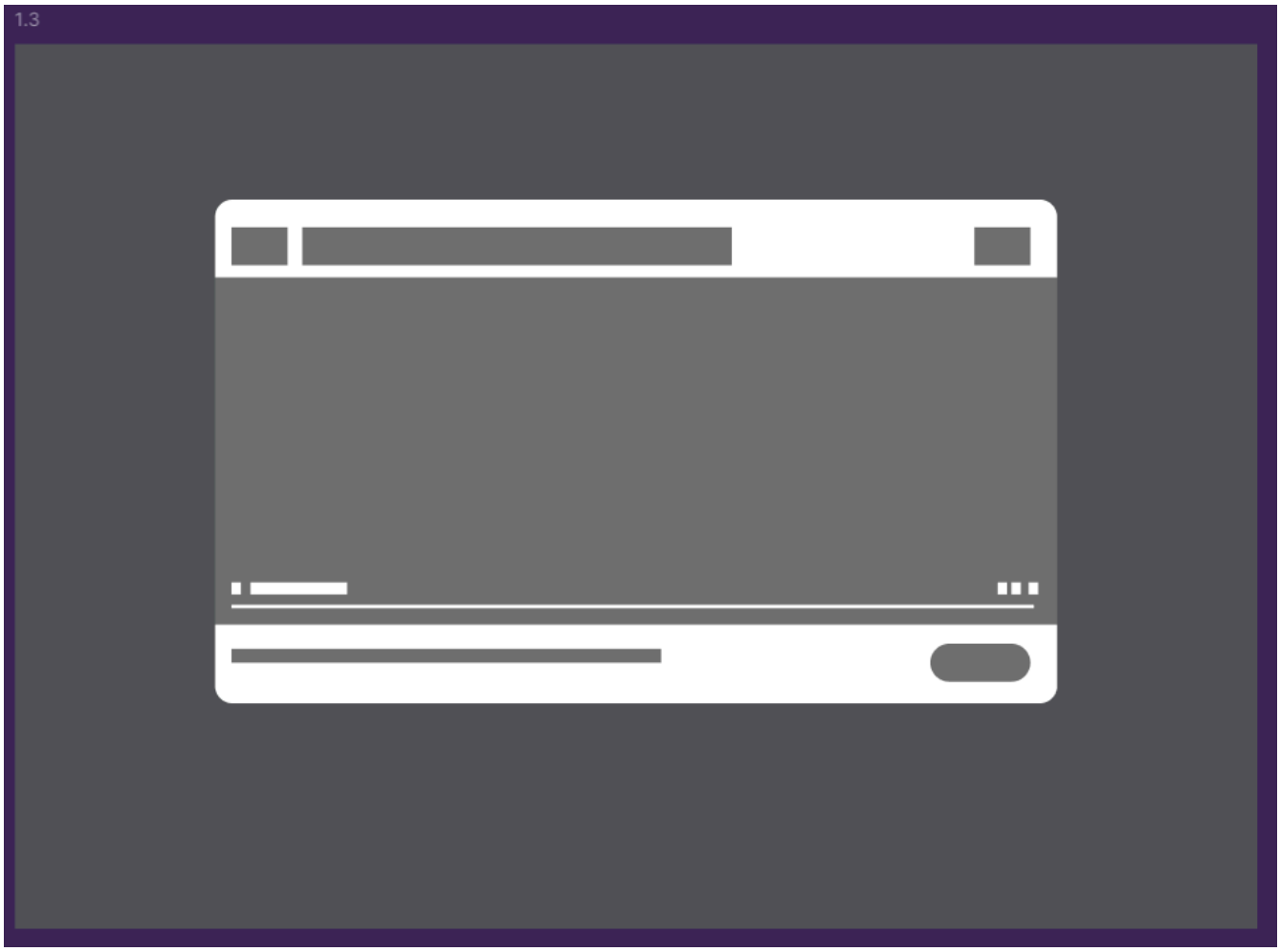


- **Use-Case Description :** Upon successful compound synthesis by the student, the system automatically displays an educational debrief overlay presenting the compound's chemical formula, bond classification (ionic or covalent), Lewis electron-dot recap, and practical real-world application. The system enforces a minimum display duration of four seconds to ensure adequate concept review while providing a visible "Continue" button. Once the minimum four-second delay has elapsed and the student clicks the "Continue" button, the overlay closes and the system unlocks the cavern barrier, allowing the student to progress to the next chamber.

- Activity Diagram



- Wireframe

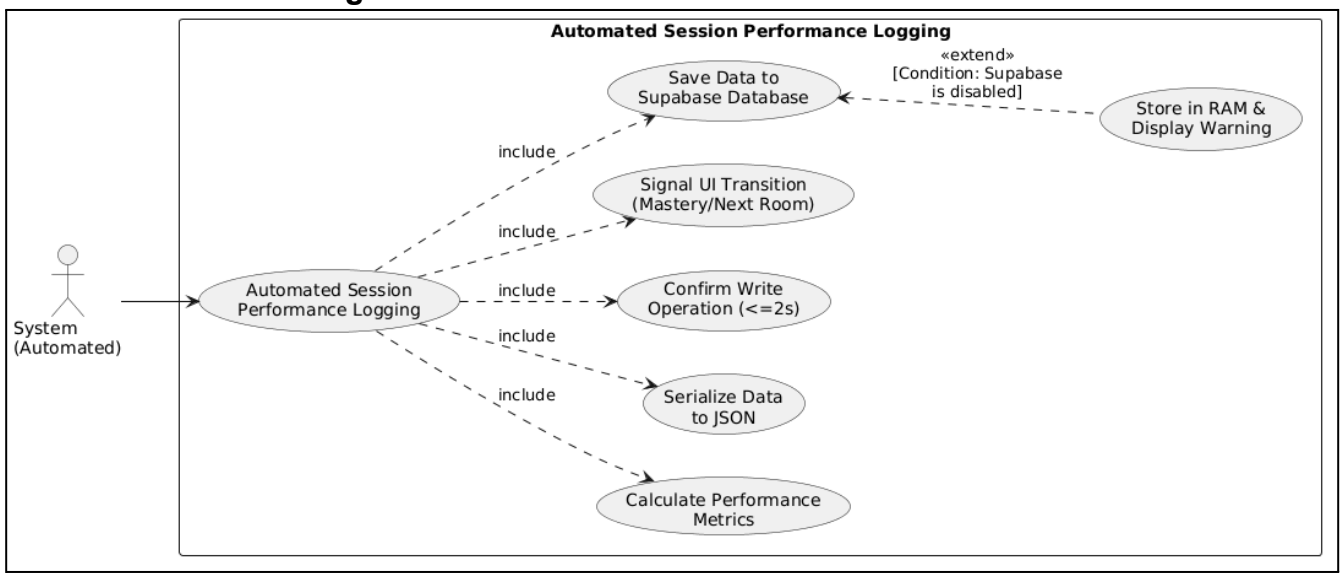


Module 2: Assessment & Progress Module

FR - 2.1 Automated Session Performance Logging

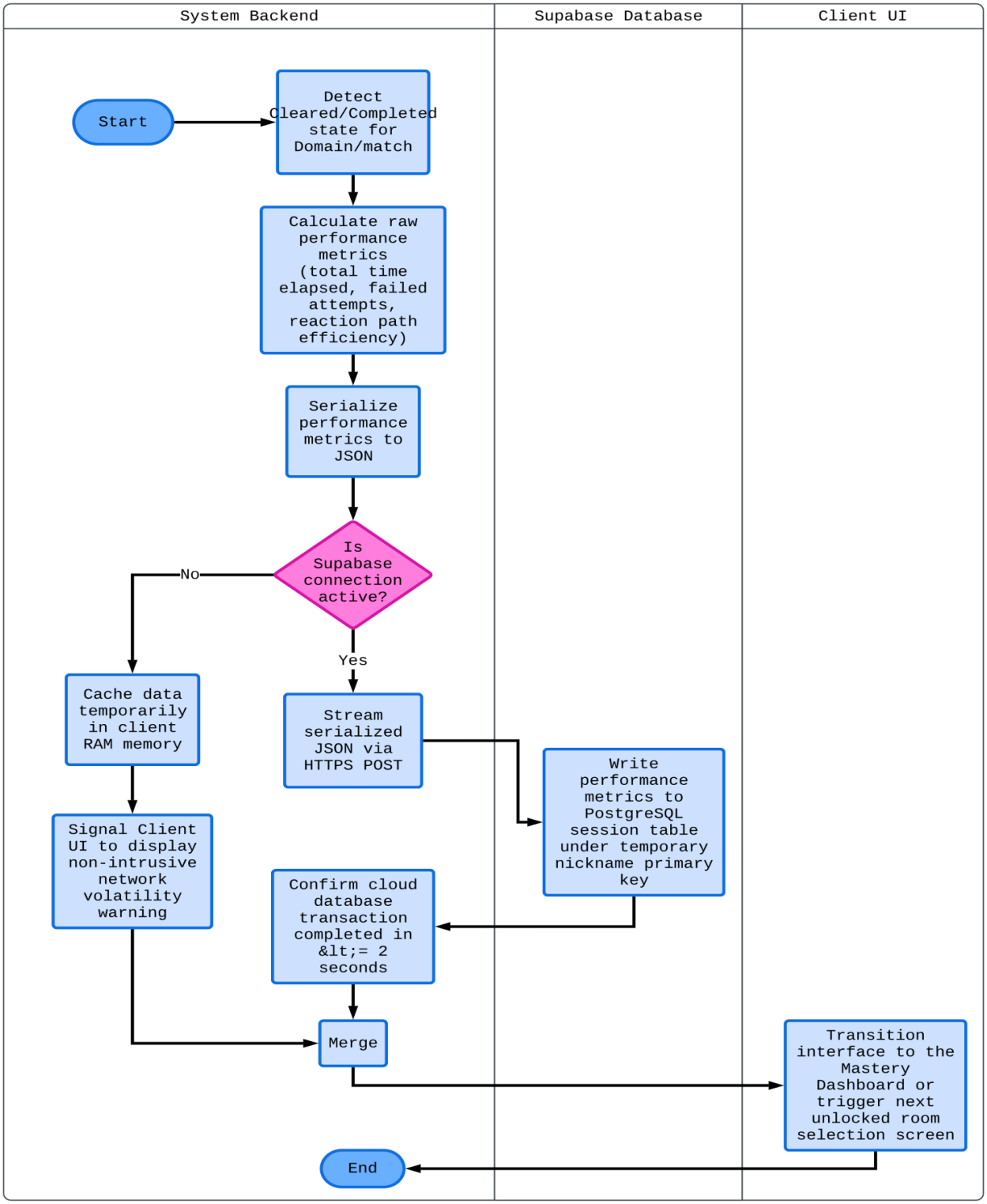
The system shall calculate accuracy and completion time upon room clearance and commit the performance record to Supabase in under 2 seconds.

- **Use Case Diagram**

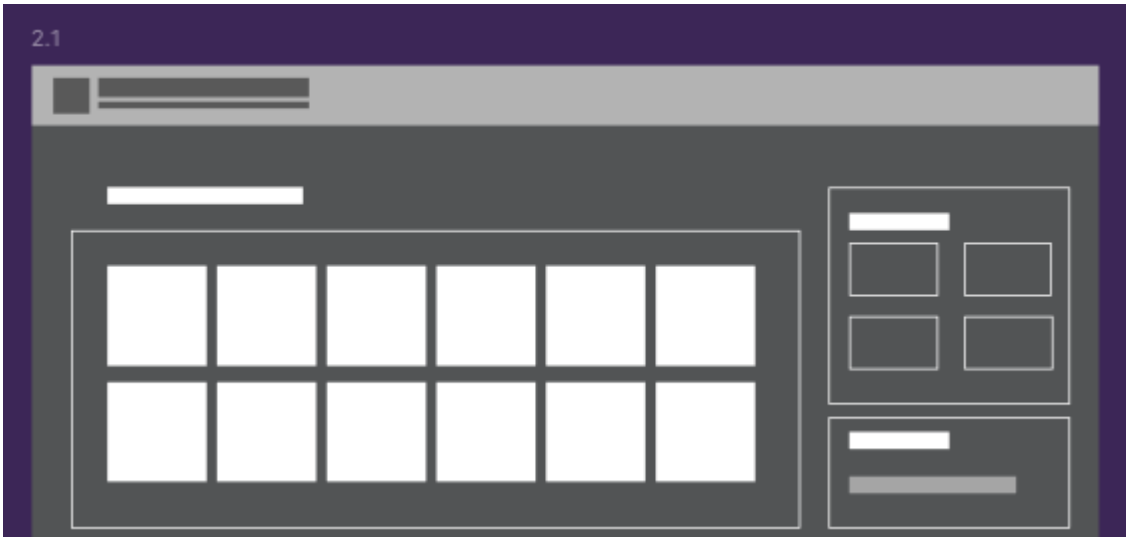


- **Use Case Description:** Upon completion of any learning Domain, the system automatically calculates the student's performance metrics based on completion speed and compound accuracy, then writes this data to Supabase under the student's temporary session nickname. The write operation is completed within less than 2 seconds. If the Supabase write fails, the system displays a user-friendly error message without crashing. The UI then transitions to the Mastery Dashboard or the next available Domain.

- **Activity Diagram: Automated Session Performance Logging**



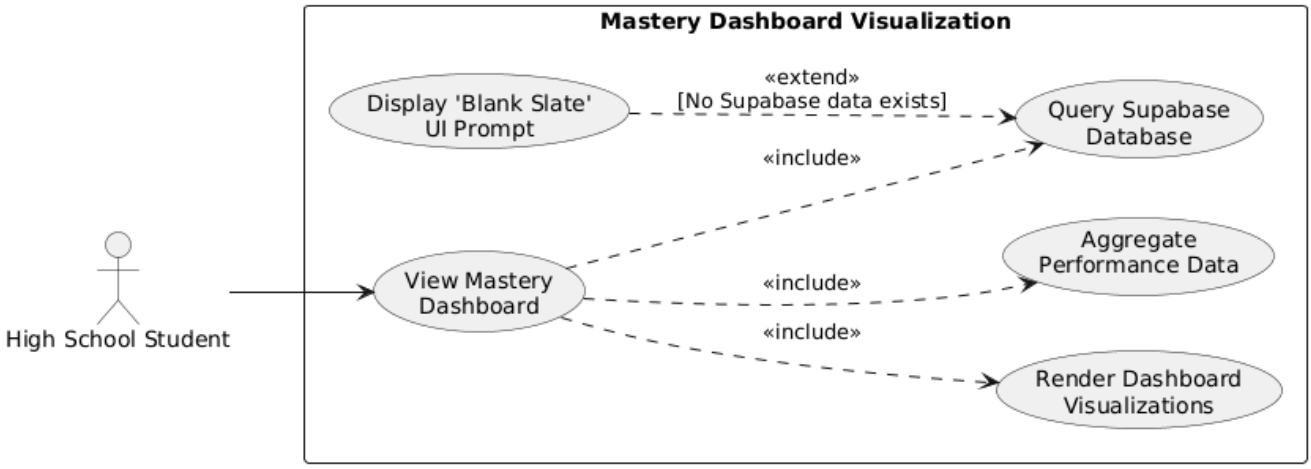
- **Wireframe**



FR - 2.2 Mastery Dashboard Visualization

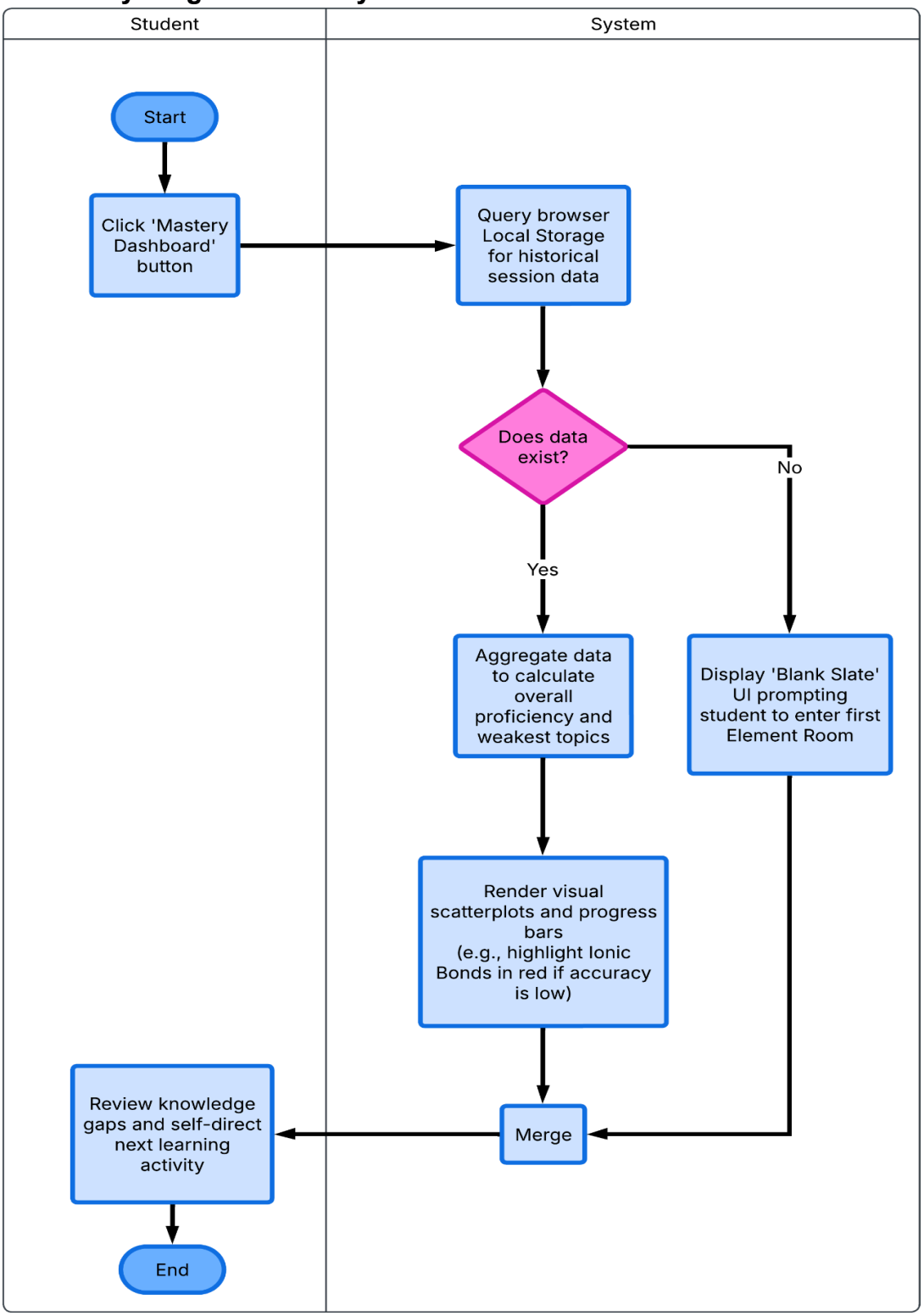
The system shall render a visual dashboard displaying accuracy percentages, completion speed, and identified weak bonding areas per domain to support self-directed review.

- **Use Case Diagram**

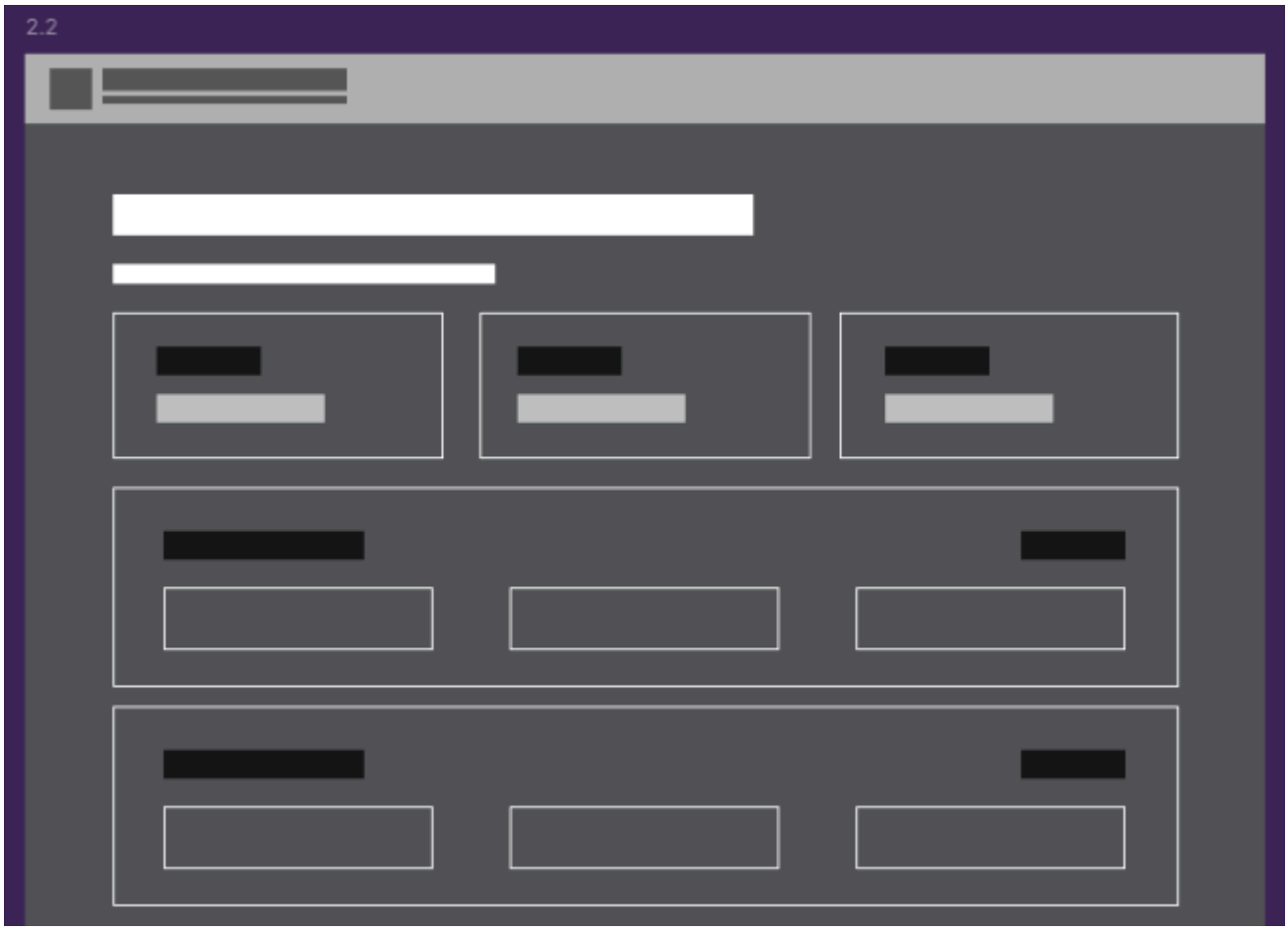


- **Use Case Description:** The student accesses their personalized Mastery Dashboard, which queries Supabase for their historical session data and renders it as visual progress indicators per Domain. If no session data exists, the system displays a prompt encouraging the student to complete their first Domain. The dashboard highlights which Domains have low accuracy, allowing the student to independently identify knowledge gaps without teacher intervention.

- **Activity Diagram: Mastery Dashboard Visualization**



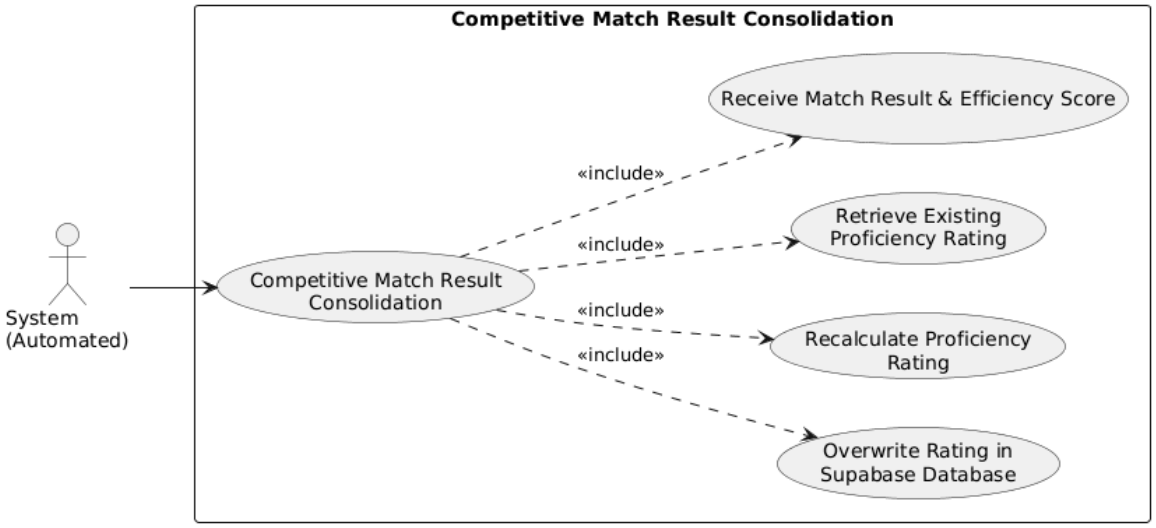
- **Wireframe**



FR - 2.3 Competitive Match Result Consolidation

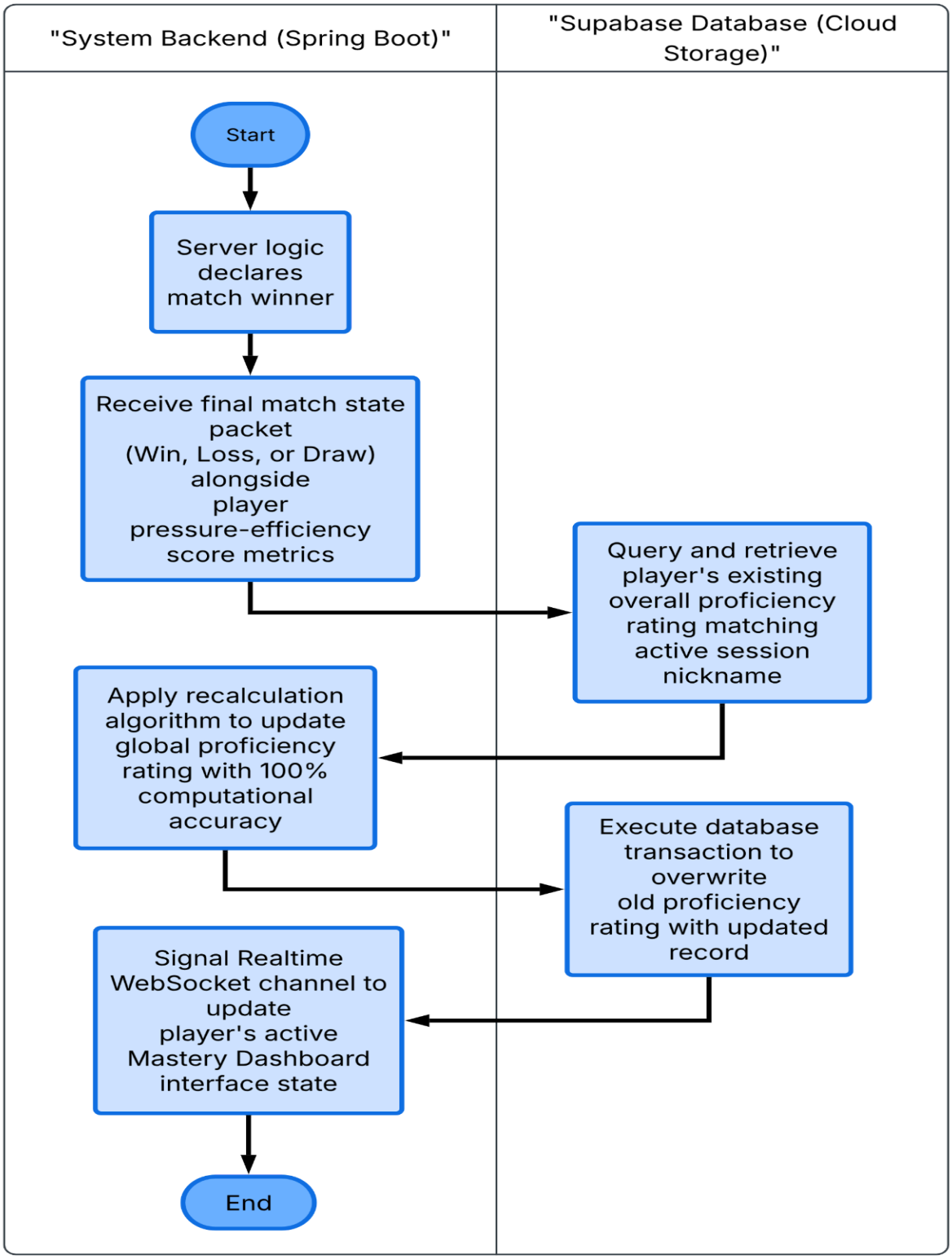
The system shall consolidate multiplayer match results (Win/Loss in multiplayer) into the students' mastery profile.

- **Use Case Diagram**

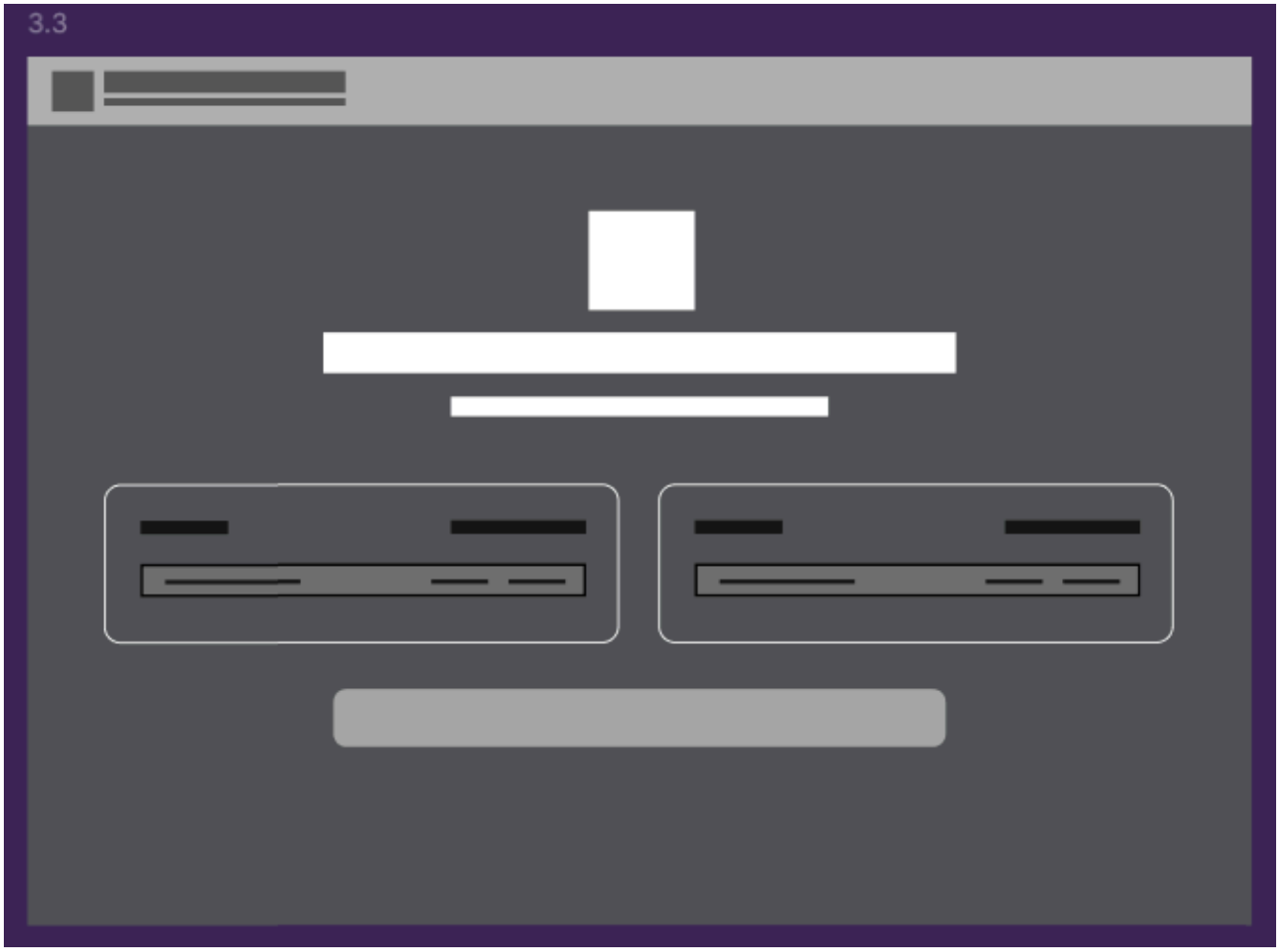


- **Use Case Description:** After a multiplayer match concludes, the system retrieves the student's existing proficiency data from Supabase, recalculates their overall rating based on the match result and Speed-to-Compound efficiency score, and overwrites the stored rating in Supabase.

- **Activity Diagram: Competitive Match Result Consolidation**



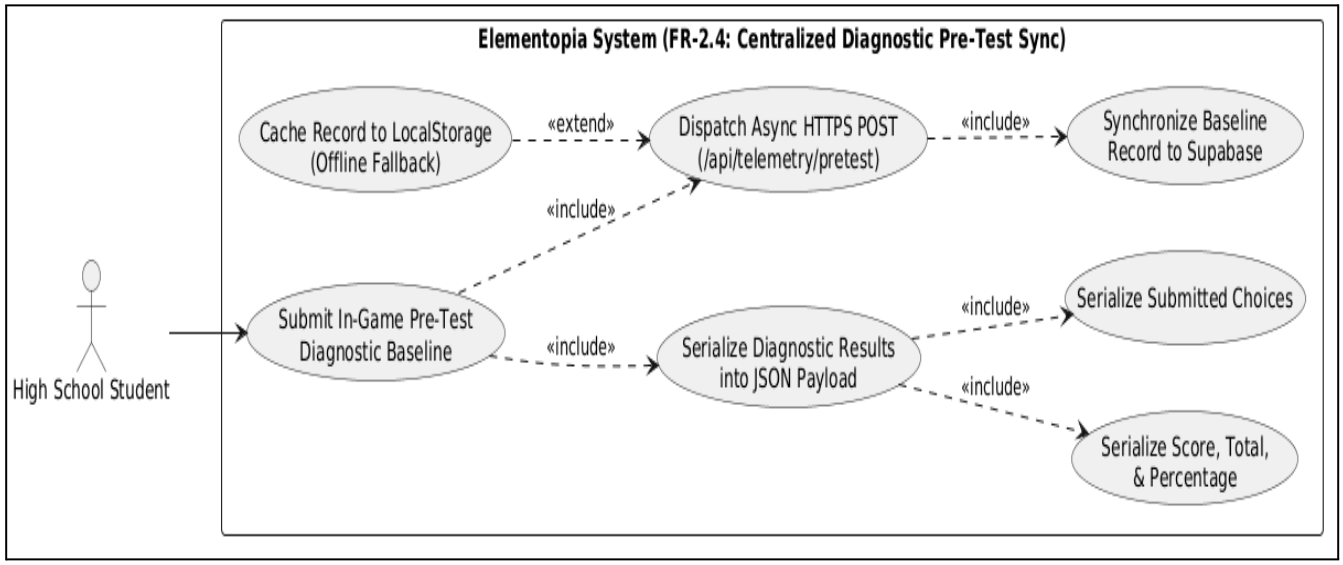
- **Wireframe**



FR-2.4: Centralized Diagnostic Pre-Test Sync

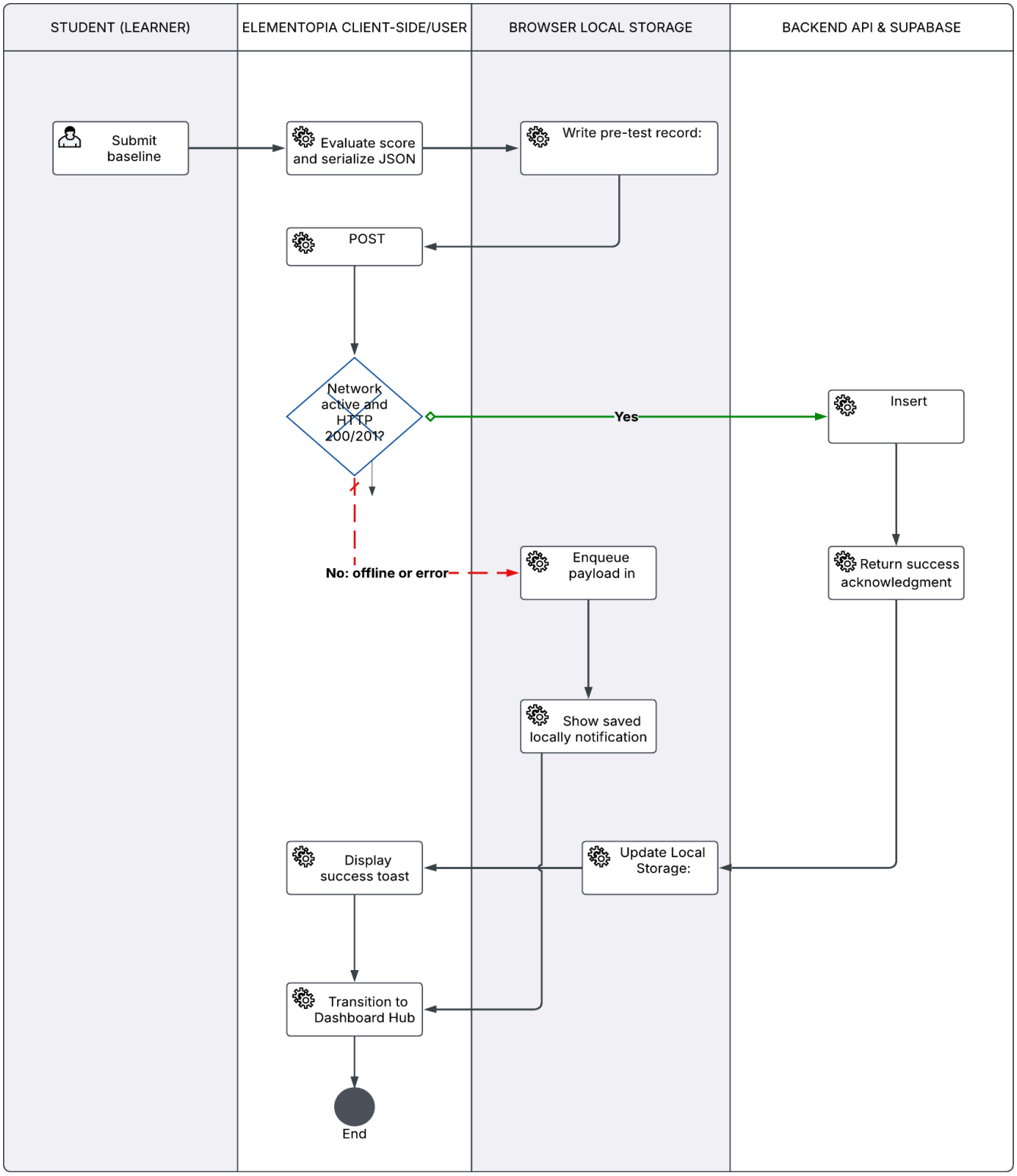
If a student submits the in-game Pre-Test diagnostic baseline, the client shall serialize the score, total, percentage, and choices into a JSON payload and dispatch an asynchronous HTTPS POST request to /api/telemetry/pretest to synchronize the baseline record to Supabase, maintaining local storage caching as an offline fallback.

- **Use-Case Diagram**



- **Use-Case Description:** Upon submission of the in-game Pre-Test diagnostic baseline, the system automatically serializes the student's diagnostic performance including their score, total items, percentage, and individual submitted choices into a structured JSON payload. The client then dispatches an asynchronous HTTPS POST request to `/api/telemetry/pretest` to synchronize the baseline record with Supabase. If the network request fails or connectivity is interrupted, the system executes an offline fallback by caching the serialized record to browser `localStorage` without disrupting gameplay. This ensures baseline diagnostic metrics are reliably preserved for subsequent performance evaluations and learning domain progression.

- Activity Diagram


STUDENT (LEARNER)
ELEMENTOPIA CLIENT-SIDE/USER
BROWSER LOCAL STORAGE
BACKEND API & SUPABASE
Submit baseline
Evaluate score and serialize JSON
Write pre-test record:
POST
Network active and HTTP 200/201?
Yes
Insert
No: offline or error
Enqueue payload in
Return success acknowledgment
Show saved locally notification
Display success toast
Update Local Storage:
Transition to Dashboard Hub
End

- Wireframe

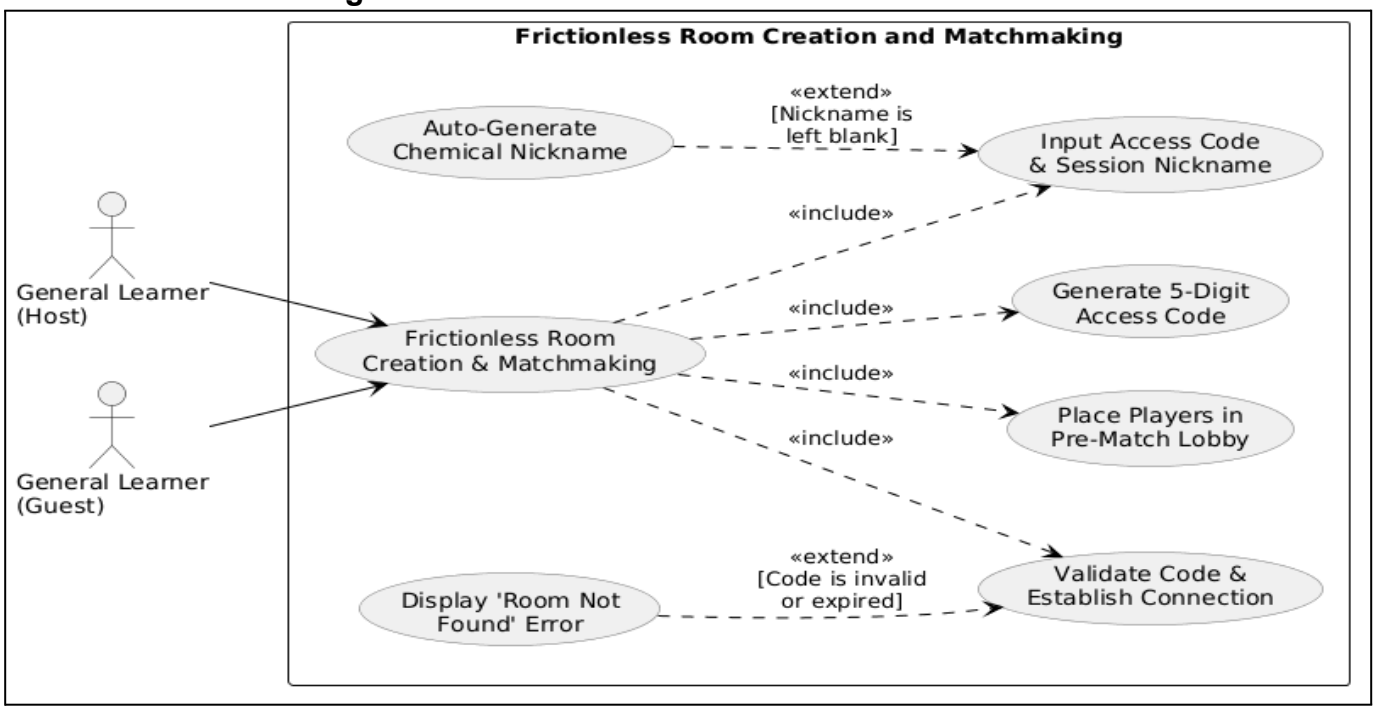


Module 3: Opponent-Based Challenge Module

FR - 3.1 Room Creation and Matchmaking

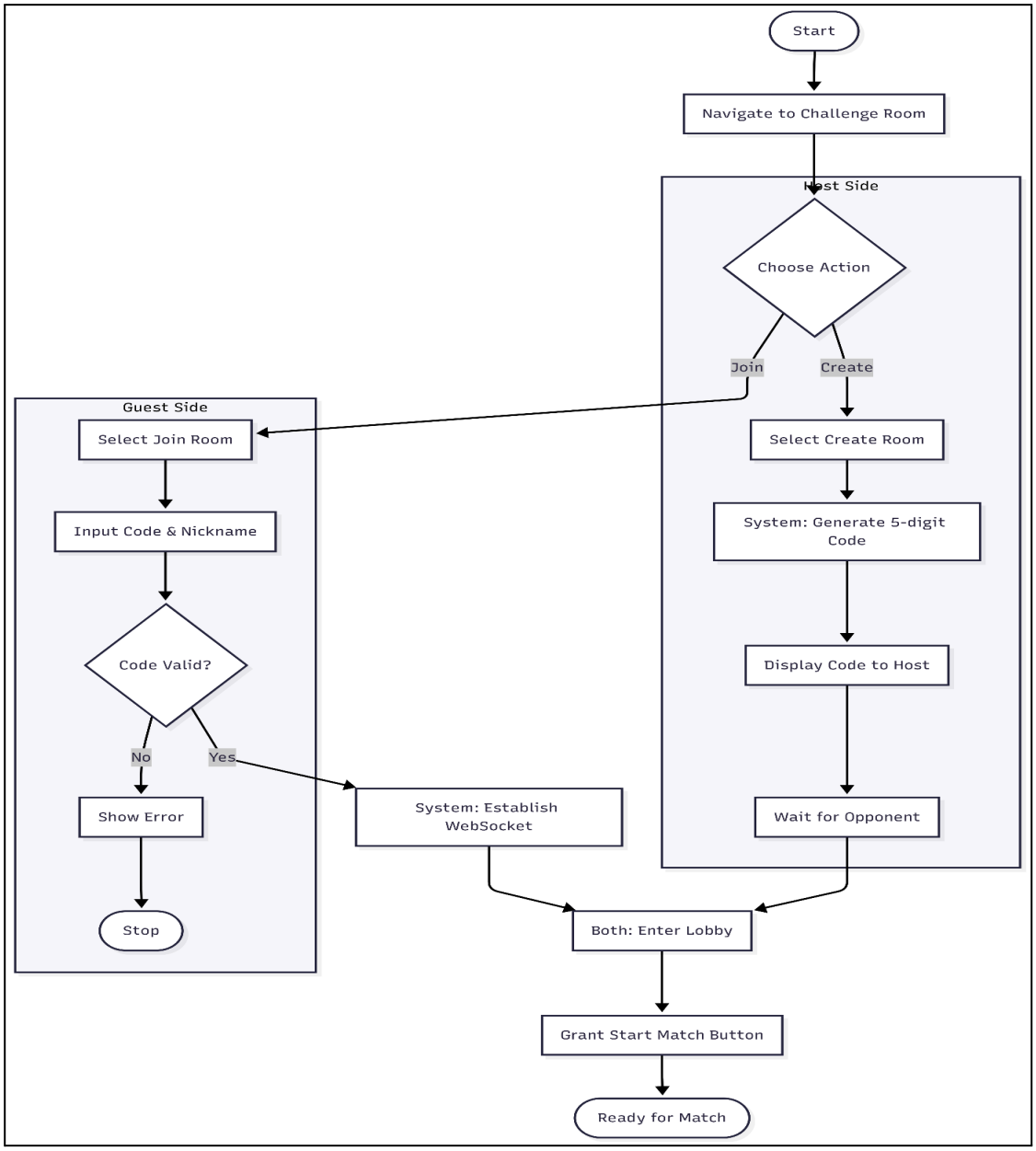
The system shall generate a unique 5-digit alphanumeric room access code for lobby creation, connecting peers via Supabase Realtime in under 5 seconds

- **Use Case Diagram**



- **Use Case Description:** A host student navigates to the Opponent-Based Challenge section and selects to create a room. The system auto-generates a 5-digit access code and displays it to the host. The host waits in a pre-match lobby. A guest student enters the same access code along with a temporary session nickname to join the room. The system validates the code, establishes a Supabase Realtime connection between both players, and places them in the pre-match lobby. If the code is invalid or expired, the system displays a room not found error. Once both players are connected, the host can start the match.

- **Activity Diagram**



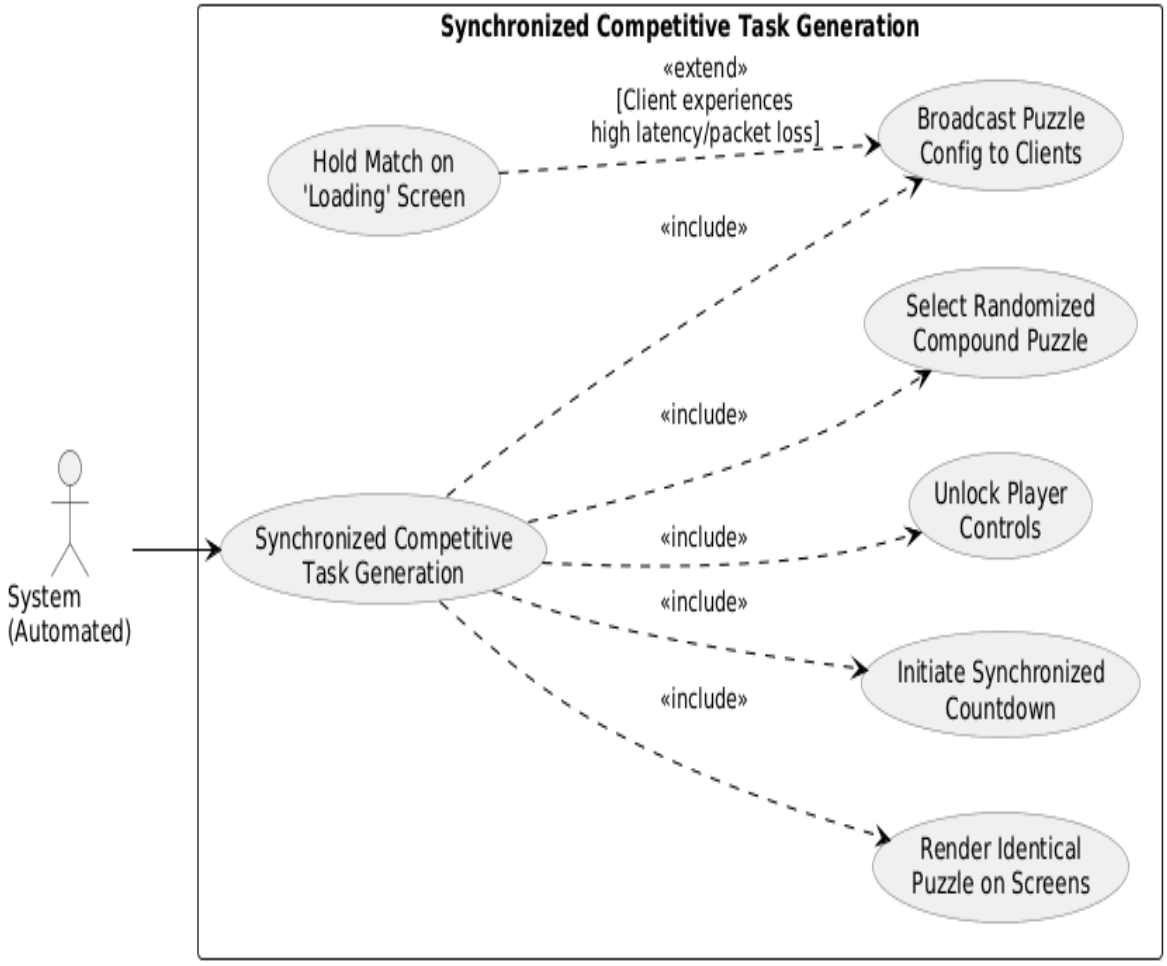
- **Wireframe**



FR - 3.2 Synchronized Competitive Task Generation

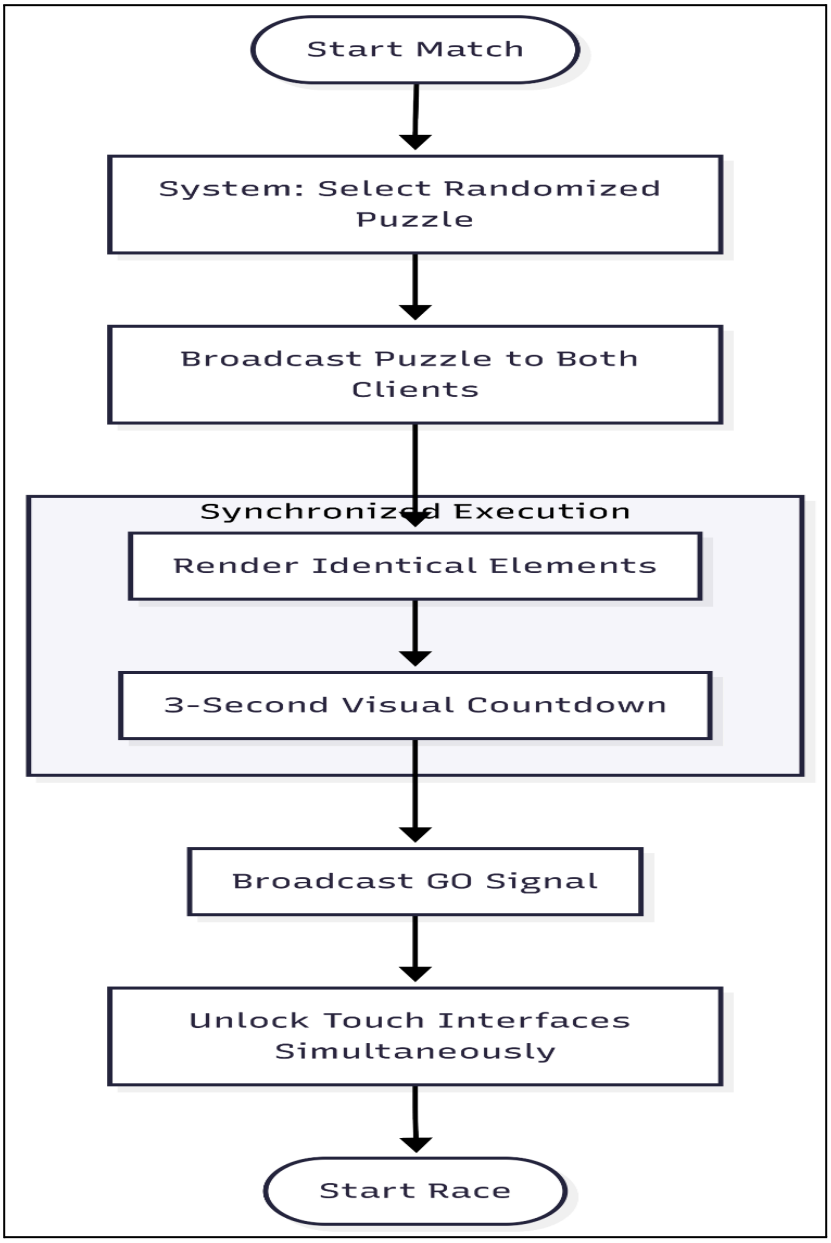
Once a match starts, the system shall broadcast an identical compound challenge and element inventory to players simultaneously with a synchronized 3-second countdown.

- **Use Case Diagram**

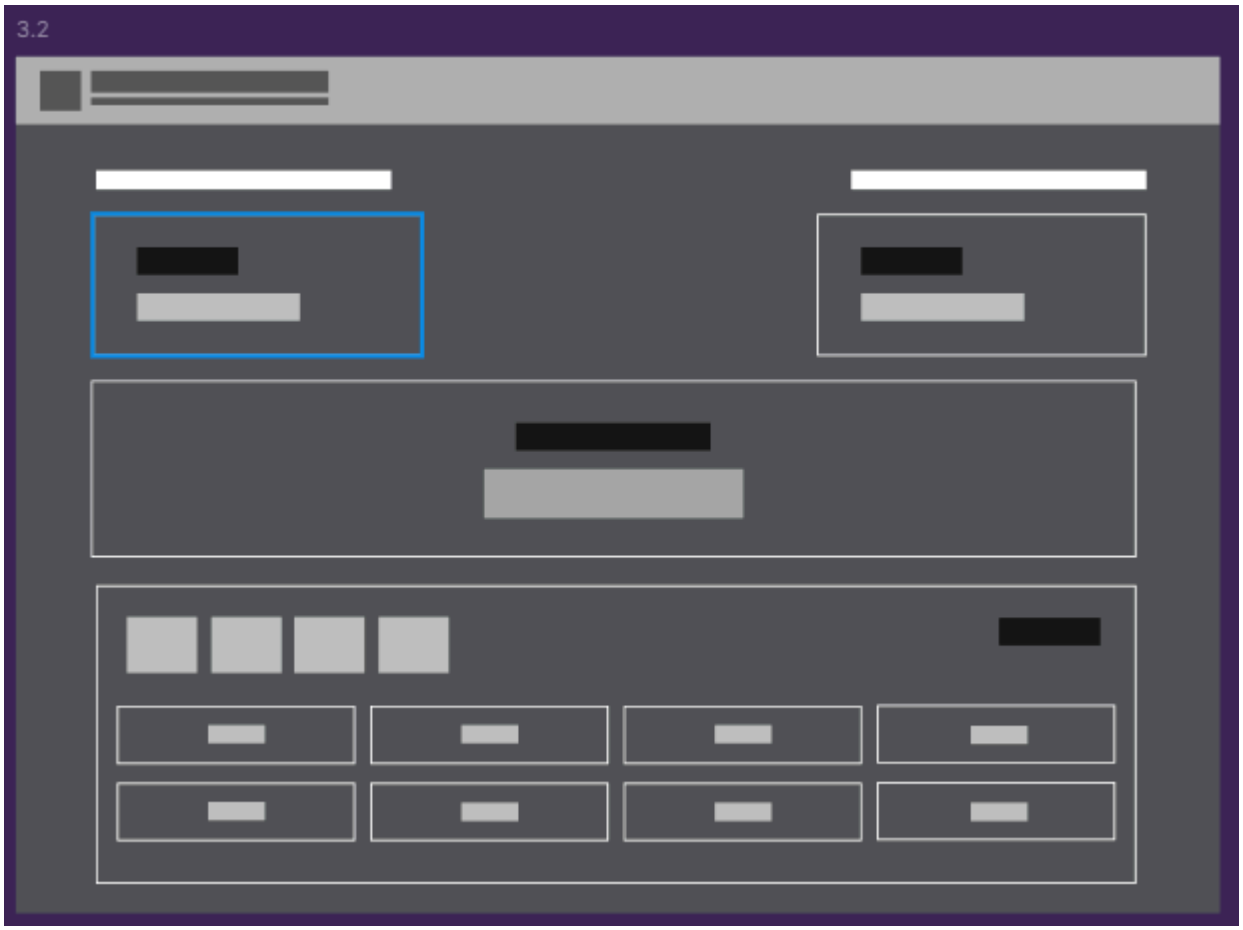


- **Use Case Description:** Once the match is initiated, the system automatically selects a randomized compound challenge from the predefined pool and broadcasts the same puzzle configuration to both connected players simultaneously via Supabase Realtime. Both players see identical element inventories and obstacles. A 3-second visual countdown is displayed before player controls are unlocked and the race begins.

- **Activity Diagram**



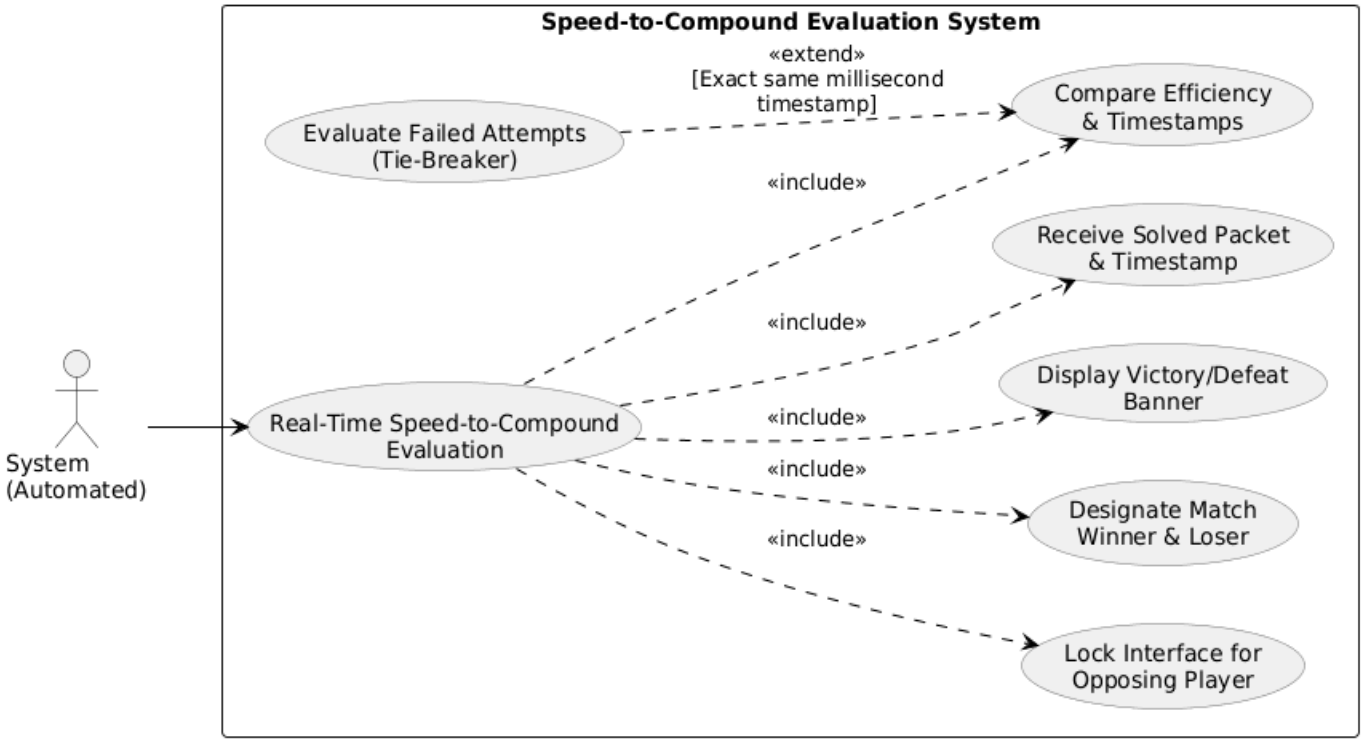
- **Wireframe**



FR - 3.3 Real-Time Speed-to-Compound Evaluation

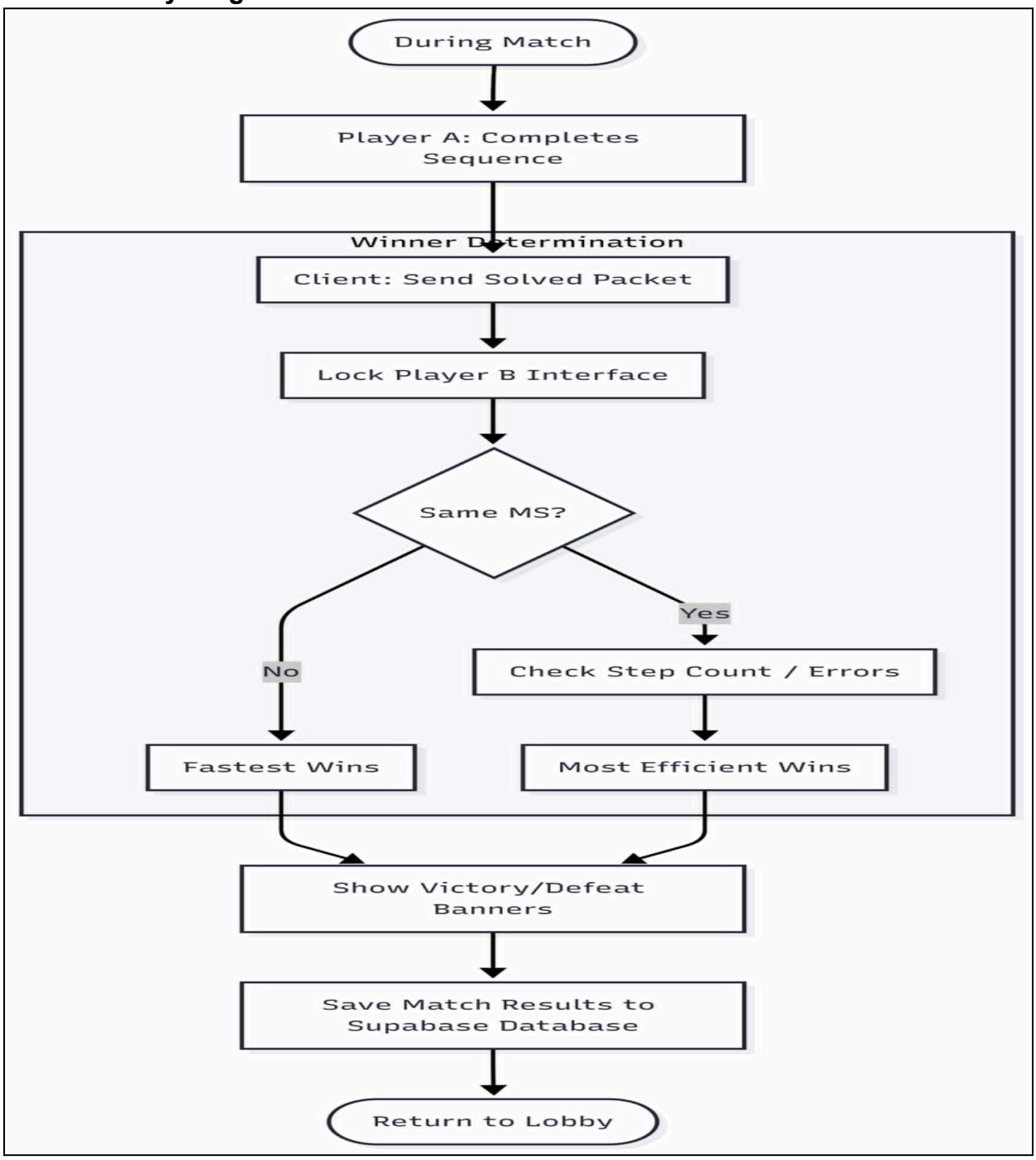
The system shall evaluate the player completion time stamps and reaction path efficiency via server-side logic, determining the match winner with accuracy.

- **Use Case Diagram**

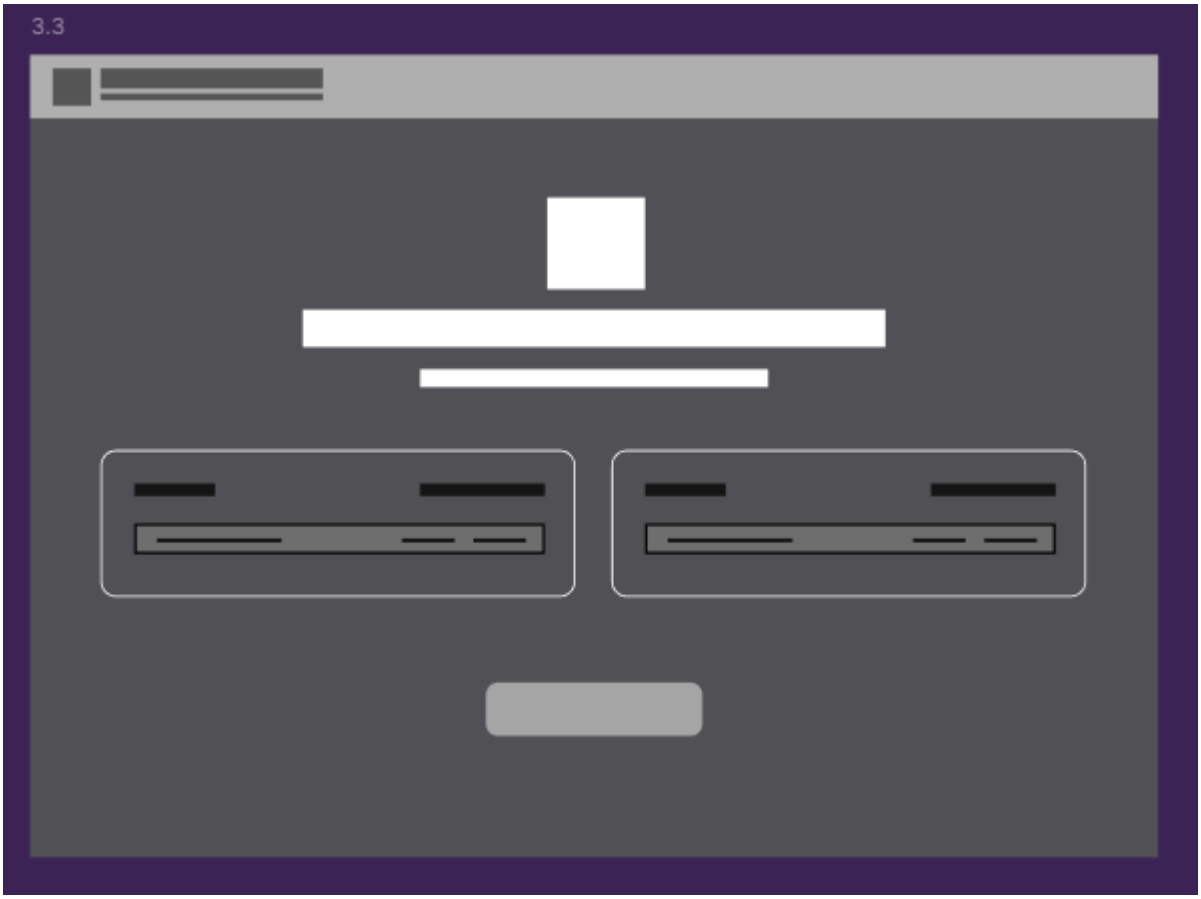


- **Use Case Description:** During the match, both players attempt to synthesize the correct compound. The first player to complete a valid synthesis sends a solved signal to the system via Supabase Realtime. The system locks the opposing player's interface, determines the winner, and displays victory and defeat banners to the respective players. If both players solve at the exact same millisecond, the player with fewer failed attempts is declared the winner. Match results and efficiency scores are saved to Supabase and reflected in both players' Mastery Dashboards.

- **Activity Diagram**



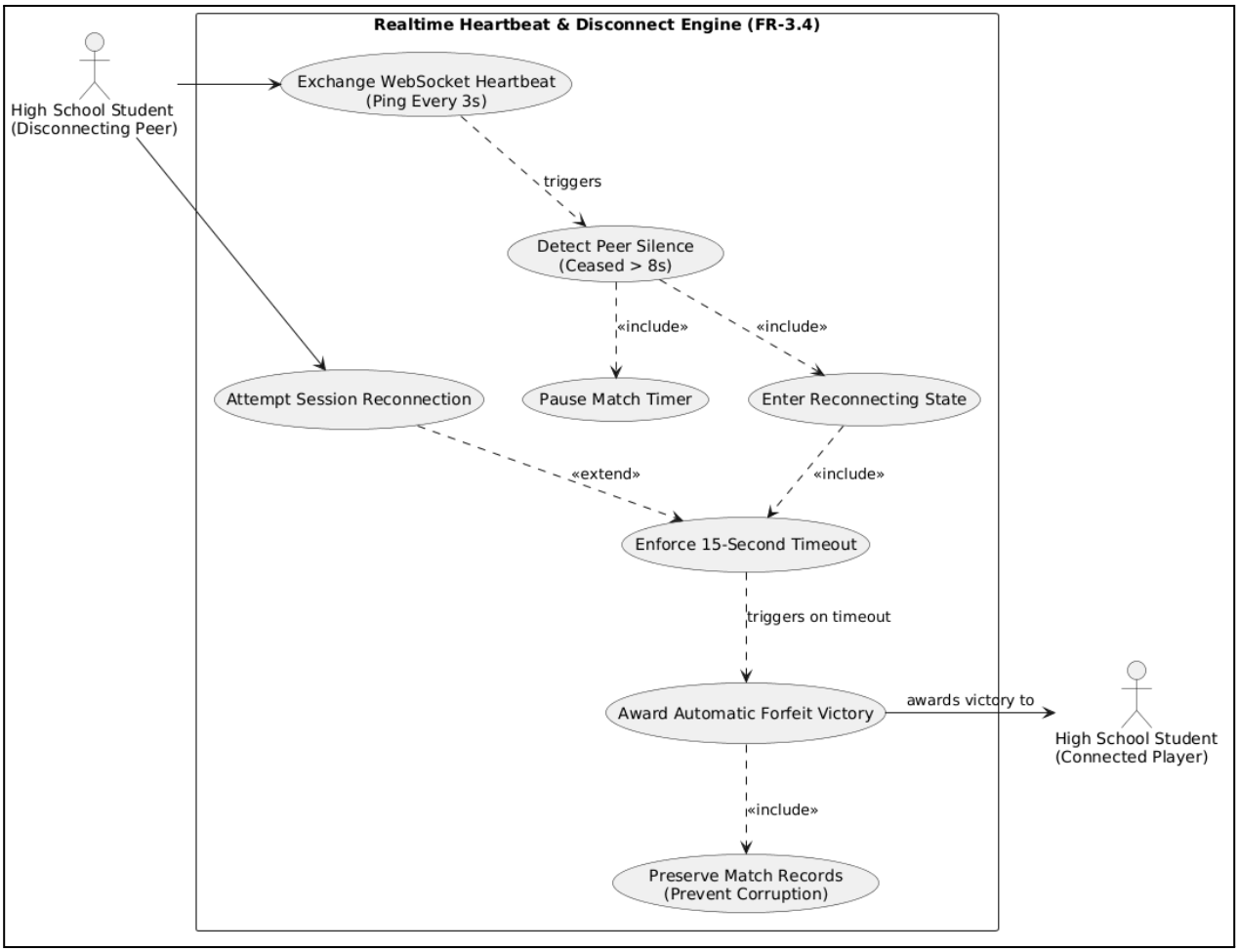
- **Wireframe**



FR-3.4 Realtime Heartbeat and Disconnect Engine

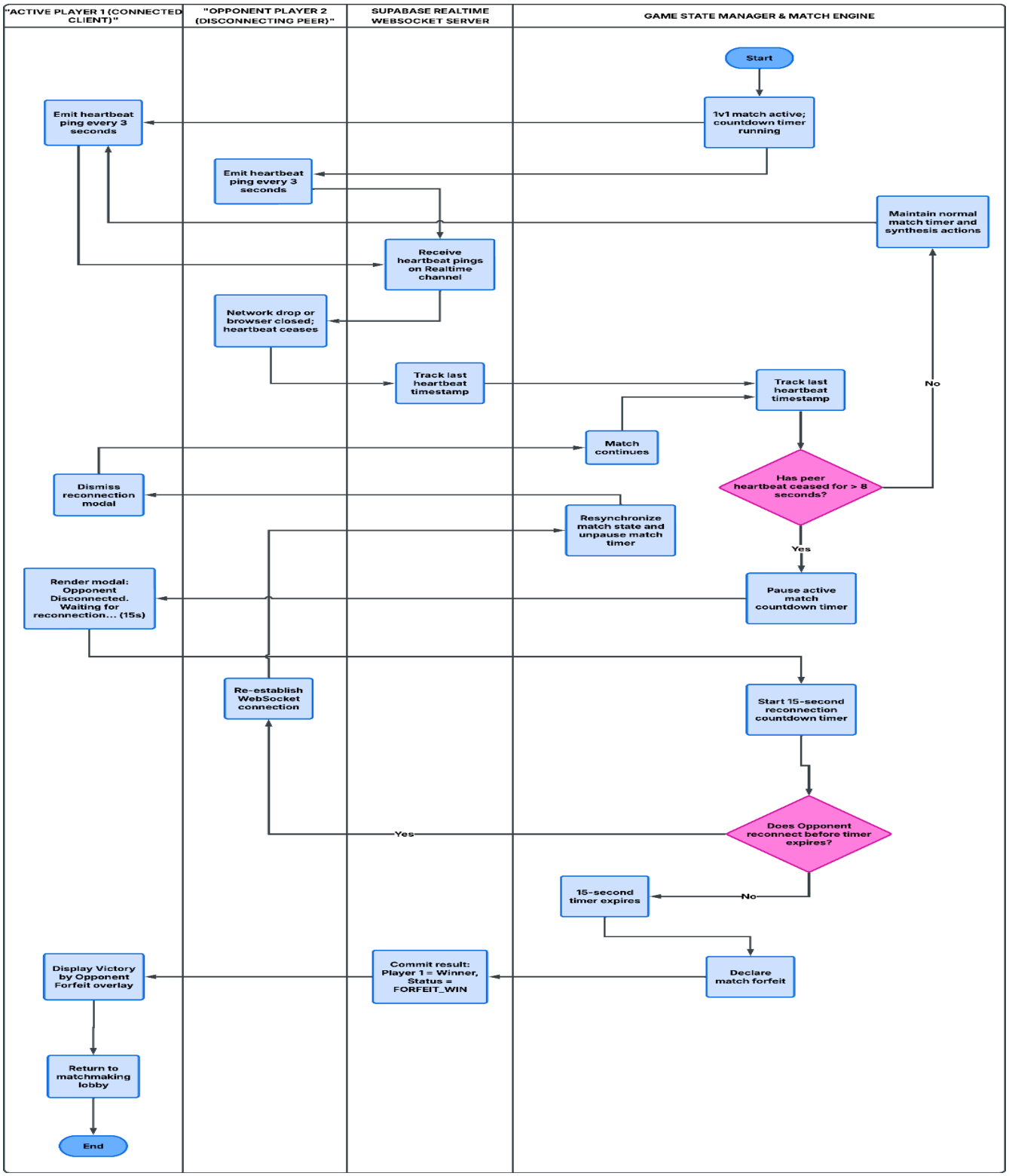
The system shall emit WebSocket heartbeat pings every 3 seconds during active matches. If a peer heartbeat ceases for more than 8 seconds, the system shall pause the match timer and enter a reconnecting state; if reconnection fails after 15 seconds, the system shall award an automatic forfeit victory to the connected player without corrupting match records.

- **Use-Case Diagram**

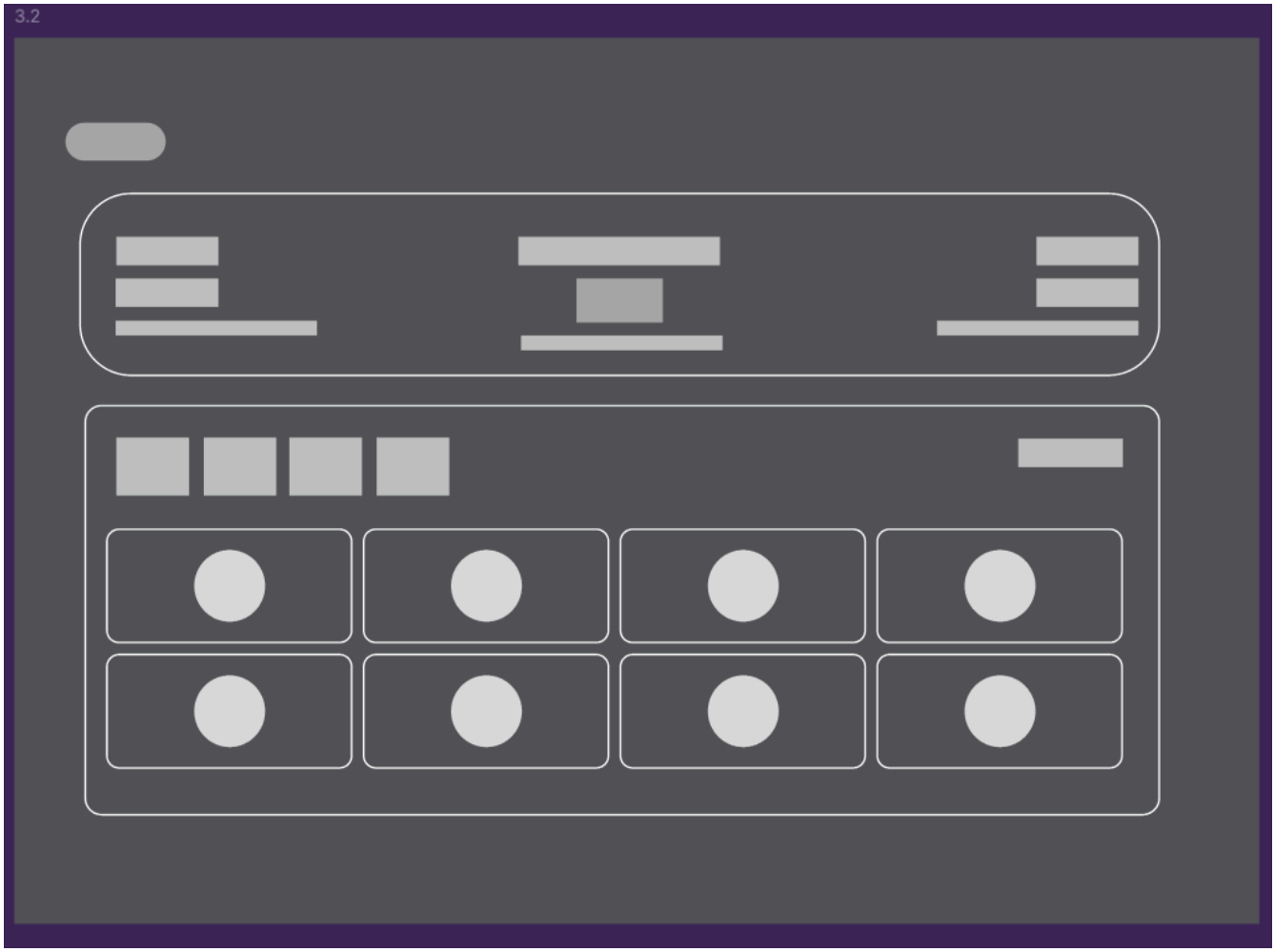


- **Use-Case Description:** During an active match, the system continuously exchanges WebSocket heartbeat pings every 3 seconds to monitor peer connectivity. If a peer's heartbeat signal ceases for more than 8 seconds, the system automatically pauses the match timer and transitions into a reconnecting state while enforcing a 15-second timeout. If the disconnecting peer fails to re-establish connection within this window, the system automatically awards a forfeit victory to the remaining connected player. The match then concludes cleanly while preserving all session records to prevent data corruption, smoothly transitioning the connected player to the post-match results screen.

- **Activity Diagram**



- Wireframe



3.4 Non-functional requirements

Performance

- Session performance data must be logged to Supabase within less than 2 seconds per Domain completion.
- The Opponent-Based Challenge Module must establish a Supabase Realtime connection in less than 5 seconds after a valid 5-digit access code is entered.
- The Mastery Dashboard must render the student's latest session data within 3 seconds of being accessed.
- Canvas drag-and-drop response latency must remain under 16ms (60 FPS rendering).
- The system must support at least 2 concurrent users per competitive room without degradation in real-time synchronization.

Security

- No personally identifiable information (PII) shall be collected. All database interactions

must execute over HTTPS/TLS 1.3.
- All communication between the frontend and Supabase must be conducted over HTTPS.
- The Opponent-Based Challenge Module must validate access codes server-side before permitting room entry.
- The system does not implement any authentication mechanism. No credentials are stored or transmitted.

Reliability

- The single-player modules must remain functional as long as Supabase is accessible.
- In the event of a Supabase connection failure during a competitive match, the system must display a user-friendly error message and terminate the session gracefully without data corruption.
- Session performance data written to Supabase must persist across browser restarts and device switches as long as the student uses the same session nickname.
- In the event of network disruption, performance and diagnostic telemetry shall be stored in an offline FIFO queue within browser localStorage and automatically flushed to Supabase upon reconnection detection (navigator.onLine).

Usability

- The interface shall enforce a minimum touch target size of 48px x 48px for all element tiles and action buttons on viewport widths between 360px and 768px, ensuring responsive mobile execution without horizontal viewport overflow.
- All puzzle interactions must be text-free, relying exclusively on visual drag-and-drop mechanics.
- The system must not require user registration or authentication to access any module.

Compatibility

- The system shall maintain cross-browser compatibility across Google Chrome v110+, Mozilla Firefox v100+, and Apple Safari v15+.
- The system must support Supabase Realtime as a required dependency for the Opponent-Based Challenge Module and session data persistence.